



FROM PRODUCTION TO POLICY: OIL & GAS — THE ROLE OF FOSSIL FUELS TODAY

Energy, Markets, and the Transition

Alan Mourgues

Petroleum Engineer

For the students of:



UNIVERSIDAD MAYOR

Facultad de Ciencias — Escuela de Geología

GEOD1023, 2025

June 2025

Alan Mourgues

MSc, Consulting Petroleum Engineer

- Hydraulic Engineer, U. de Chile
- MSc Petroleum Engineering, Imperial College, U. of London
- Petroleum/Reservoir Engineering Consultant, based in Australia (moREc)
- Founder of CrowdField
- Co-founder of GeoDataX



crowdfield.net



geodatax.tech

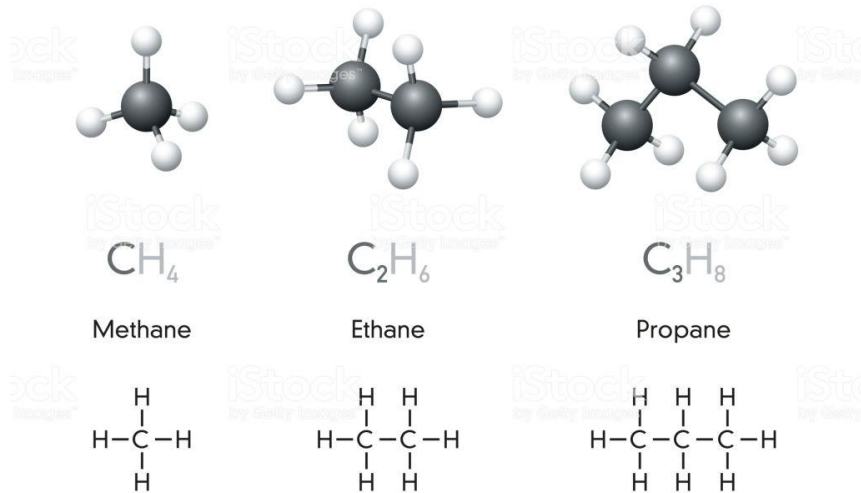
WHAT WE'LL COVER

- How Oil & Gas are produced and traded
- What affects prices
- Oil vs. gas markets
- Oil & Energy demand trends
- The Energy Transition

NOT ALL CRUDE OIL IS THE SAME



Alkanes : C_nH_{2n+2}



Name	Molecular Formula
Methane	CH_4
Ethane	C_2H_6
Propane	C_3H_8
Butane	C_4H_{10}
Pentane	C_5H_{12}
Hexane	C_6H_{14}
Heptane	C_7H_{16}
Octane	C_8H_{18}
Nonane	C_9H_{20}
Decane	$C_{10}H_{22}$

Nat Gas

LPGs

Condensate

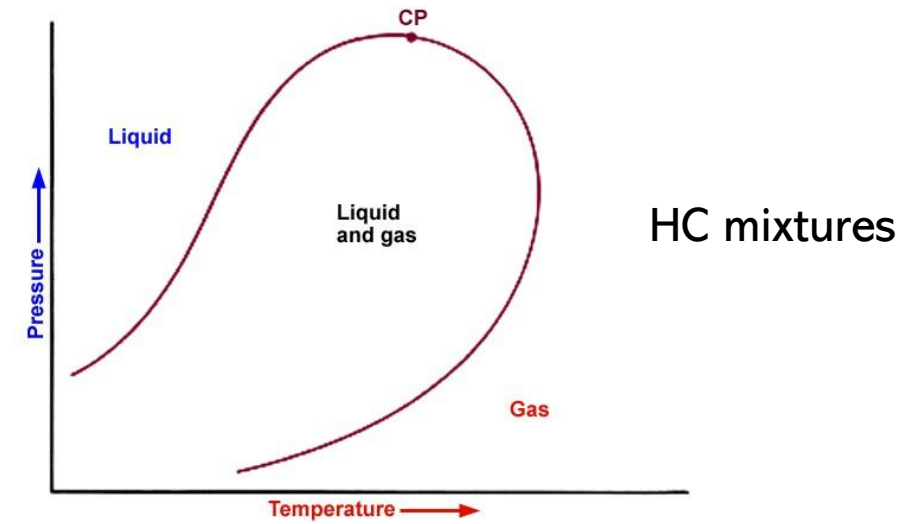
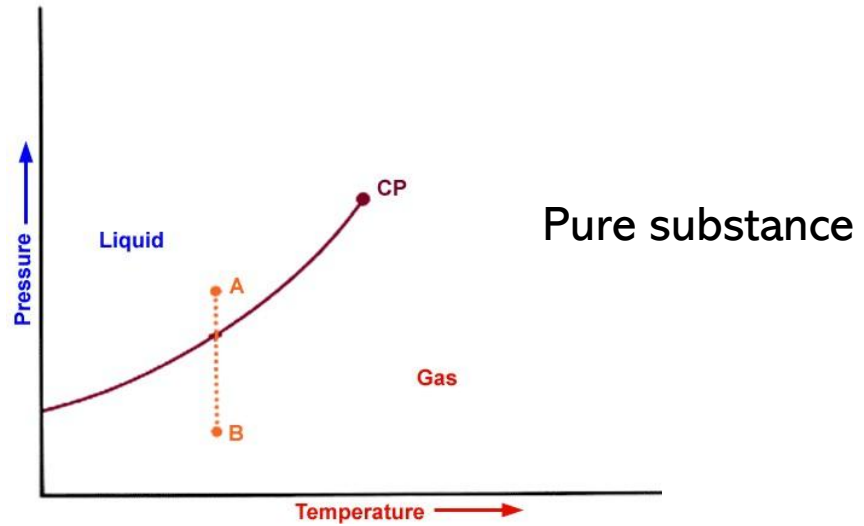
volatile

heavy

■ Impurities:

- Sulfur, Nitrogen, CO₂, etc.

PHASE BEHAVIOUR



OIL BENCHMARKS: NOT ALL CRUDE OIL IS THE SAME

- Benchmarks: 'reference oils'
 - **Brent** – North Sea; light and sweet.
 - **WTI** (West Texas Intermediate) – USA; lighter and sweeter than Brent
 - **Dubai/Oman** – Middle East; heavier, more sour; used in Asia

→ Price of other oils is set as a differential to these benchmarks

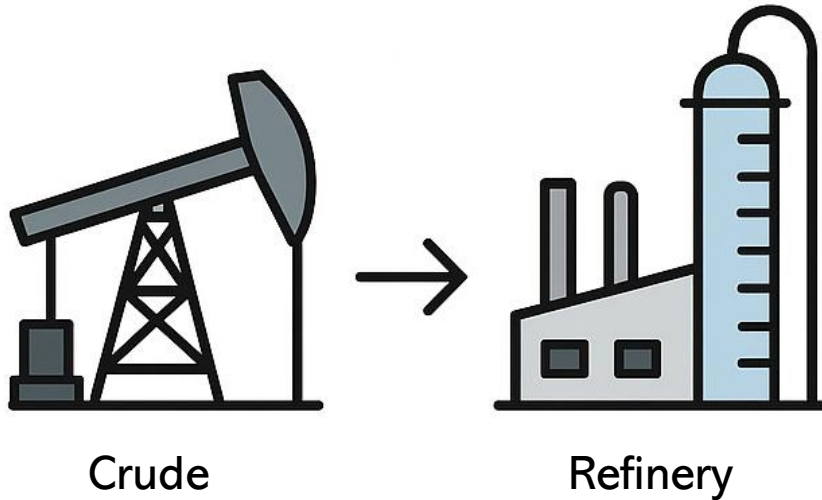
Sweet: low sulfur (Easier and cheaper to refine, cleaner product yield)

Sour: high sulfur (Higher cost and energy)

Sulfur, H₂S (hydrogen sulfide), SO₂ (sulfur dioxide)

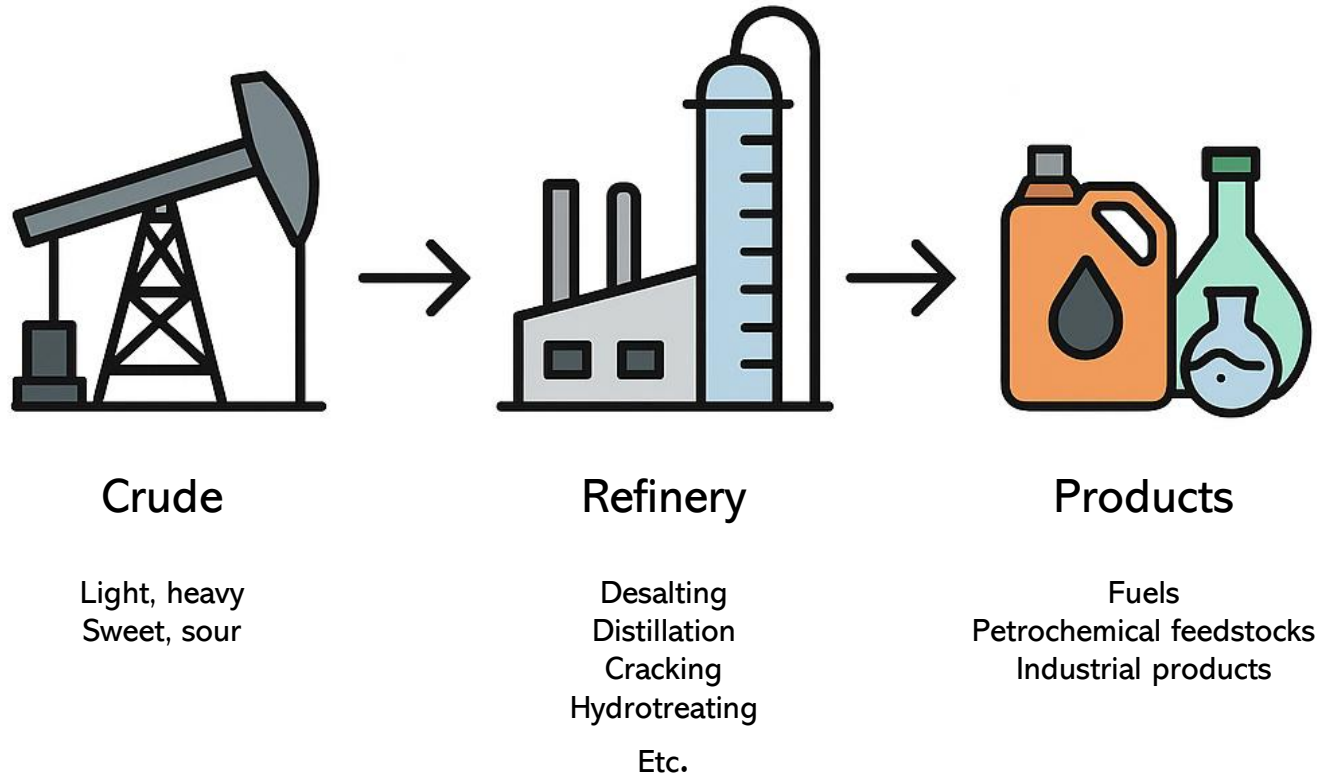
→ corrosion, pollution, poison, toxic gases

WHO BUYS CRUDE?



- **Refiners (Direct Buyers)**
 - Integrated Oil Companies (eg ExxonMobil, Shell, BP)
 - Independent refiners
- **Commodity Traders (middlemen)**
 - Price arbitrage, storage, logistics, long-term contracts
- **NOCs (Saudi Aramco, Petrobras, PDVSA)**
 - Sell to refiners
 - Swap crude for refined products
- **Gov's & strategic reserves**
 - Emergency stockpiles

PETROLEUM VALUE CHAIN



■ Fuels:

- **Gasoline** (cars), **Diesel** (trucks, buses, trains, generators), **Kerosene** (jet fuel), airplanes, **LPG**, cooking, heating, some vehicles

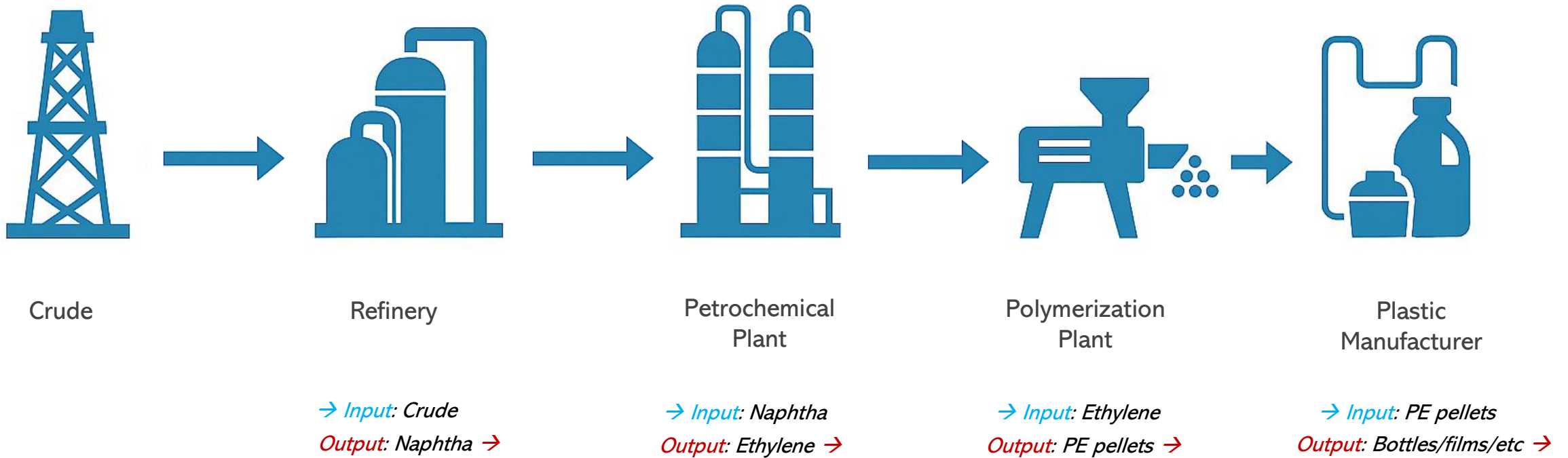
■ Feedstocks:

- Ex: naphtha → ethylene/propylene/butadiene
- End products:
 - **Polyethylene** → plastic bags, bottles, containers
 - **Polypropylene** → car parts, medical equipment, textiles
 - Etc.

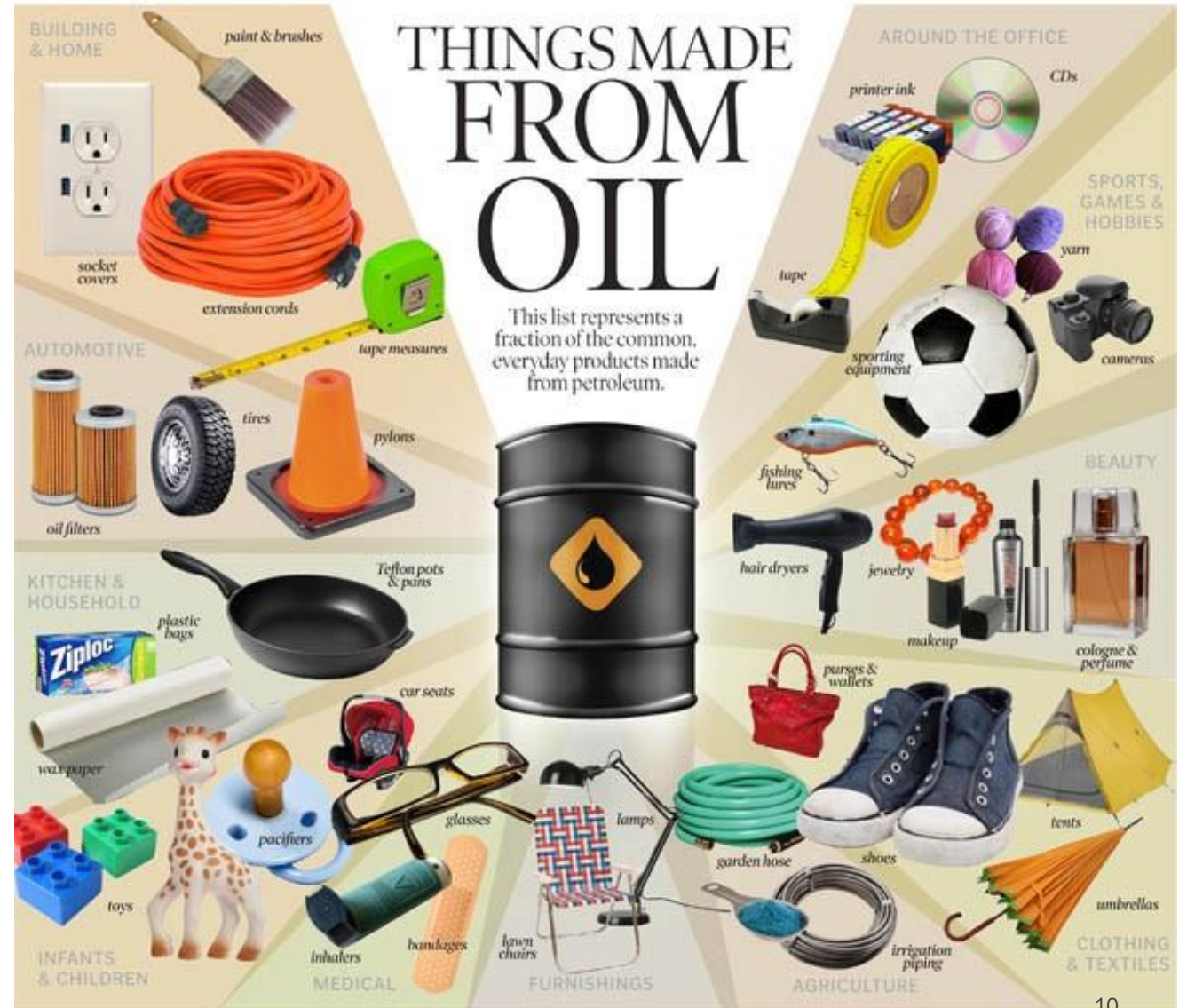
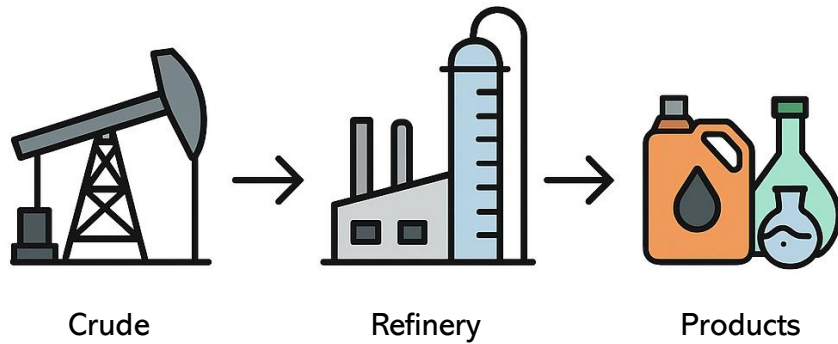
■ Industrial Products:

- Asphalt / Bitumen: roads, roofing
- Lubricants: engines, machinery
- Waxes: candles, cosmetics
- Sulfur: fertilizers, chemicals

CRUDE OIL TO END PRODUCT – EX: PLASTICS



PETROLEUM PRODUCTS



OIL: THE WORLD'S MOST TRADED COMMODITY

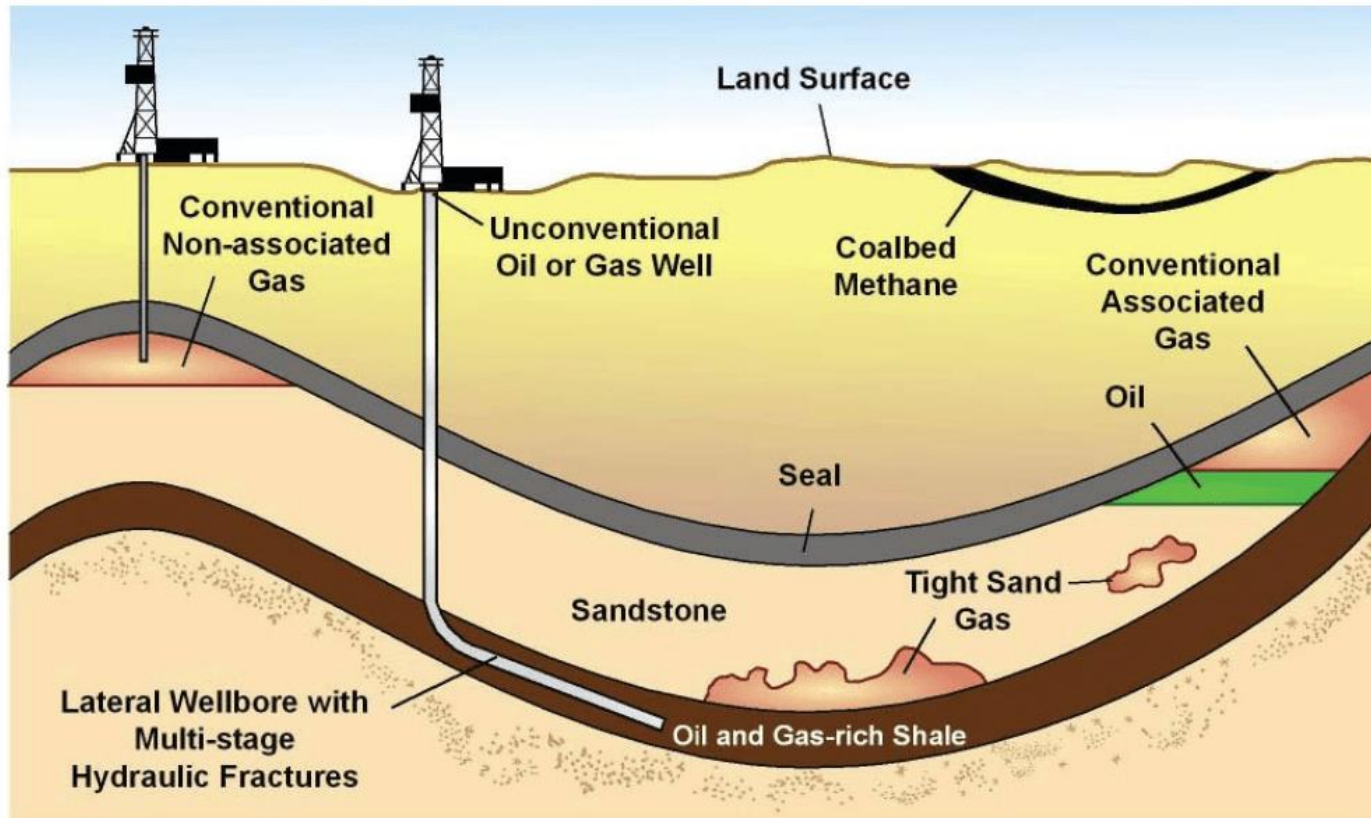
Why is Oil a Commodity?

- Fungible
 - One barrel of Brent is nearly identical to another of the same grade
- Standardized
 - Global benchmarks: Brent, WTI, Dubai, etc.
- Traded in high volumes
 - Traded daily on the spot and futures markets
- Universal demand
 - It powers everything

Other commodities:

- *Natural gas, coal*
- *Gold, silver*
- *Copper, Aluminium*
- *Wheat, Corn*
- *Coffee, Sugar*
- *Etc.*

CONVENTIONAL VS UNCONVENTIONAL



Source: EIA

Conventional

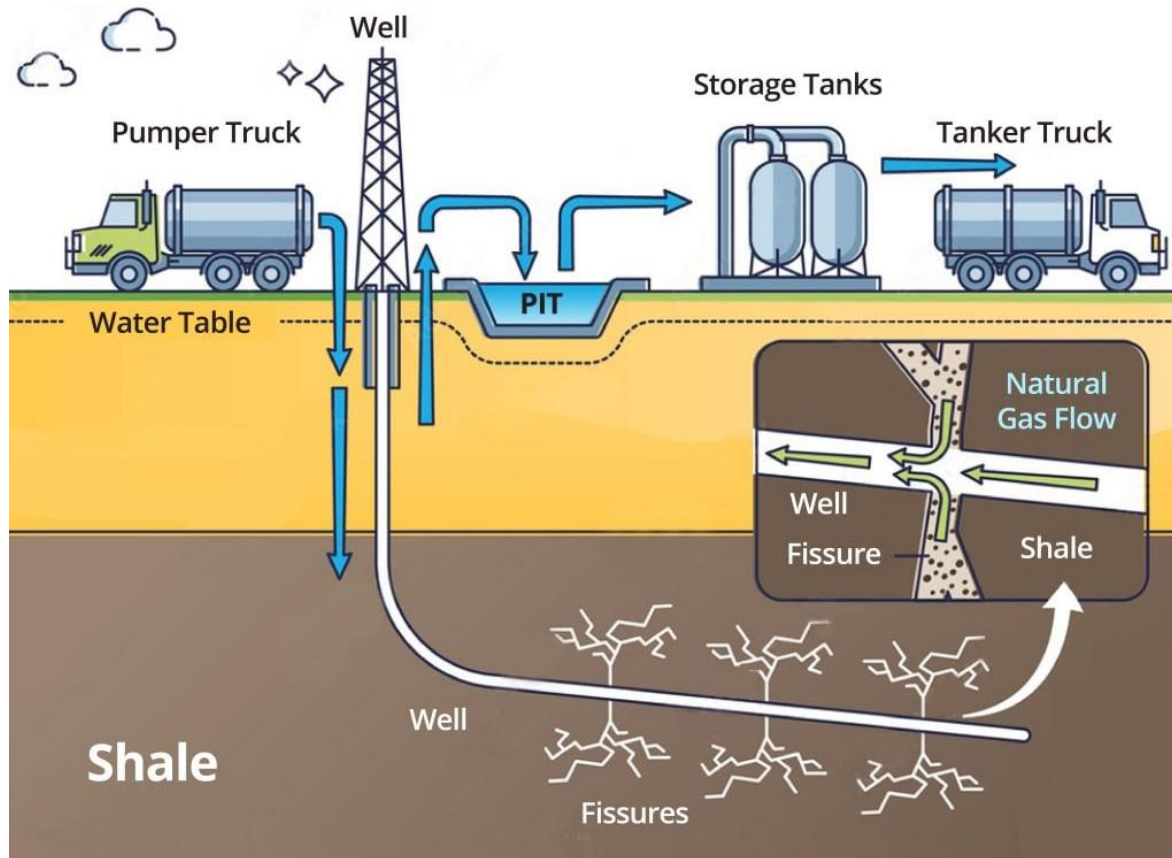
- Discrete accumulations in porous, permeable rocks
- Controlled by natural pressure and traps
- Extracted easily with minimal processing

Unconventional Resources

- Widespread, low-permeability formations
- Not trap-dependent; minimal hydrodynamic influence
- Require stimulation and advanced processing (e.g., fracking)

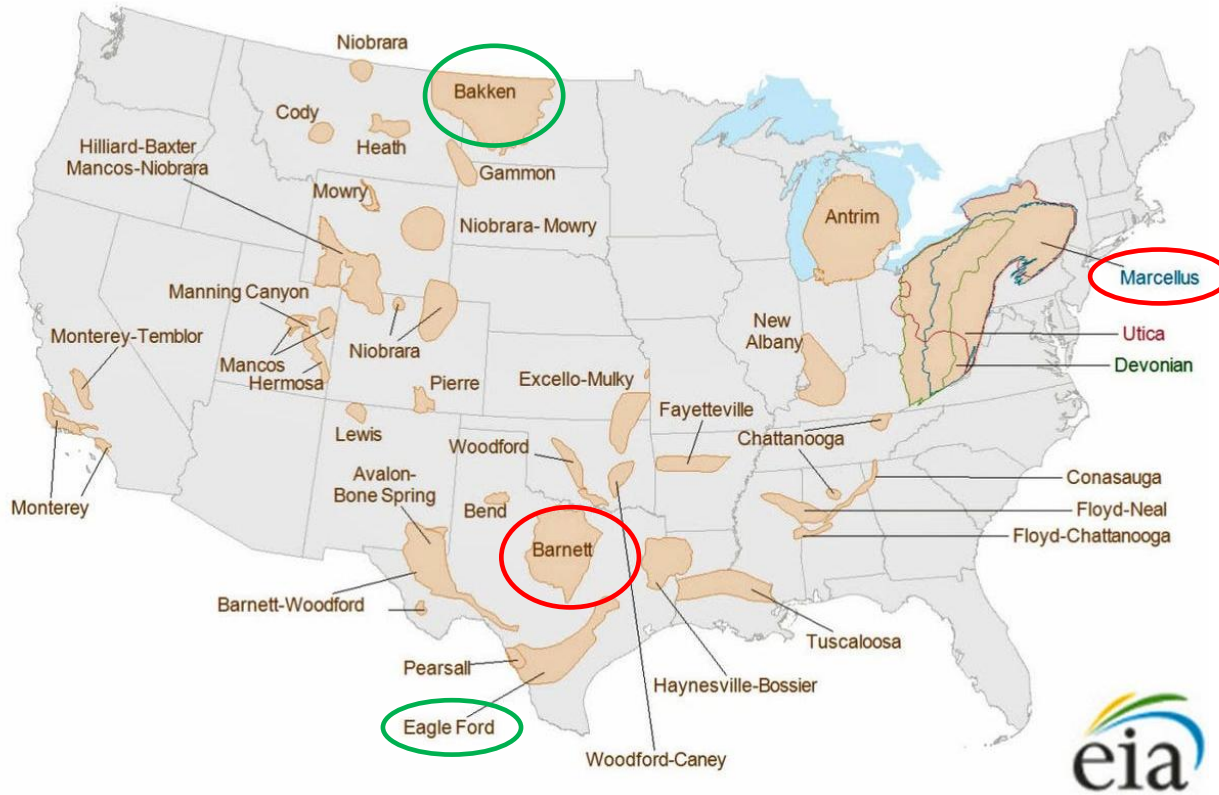
Shale oil
Tight gas
CBM / CSG
Others.

HYDRAULIC FRACTURING (FRACKING)



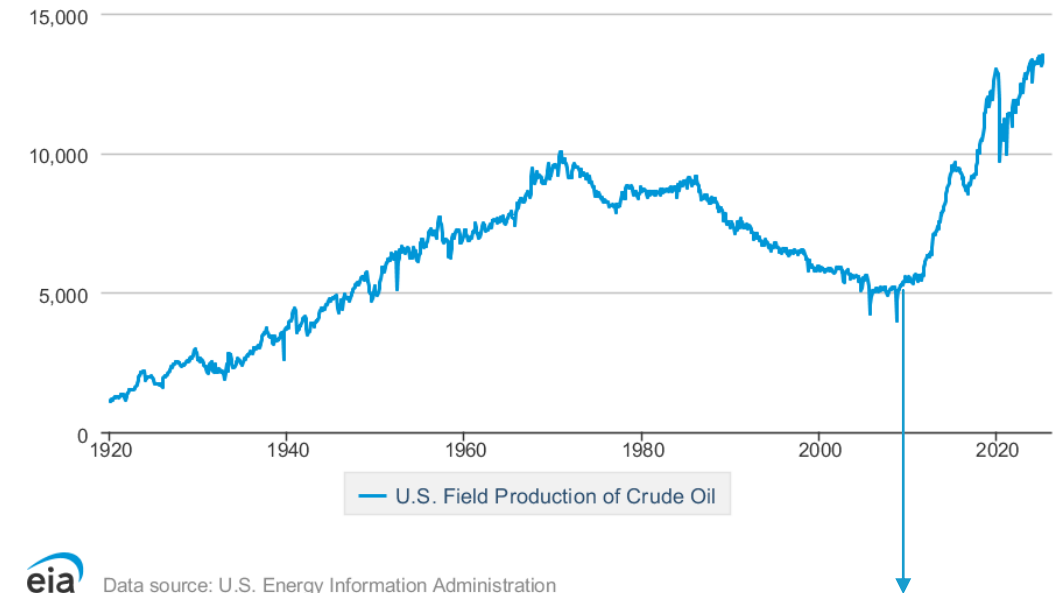
- Water, sand, and chemicals are pumped underground
- High pressure fractures the tight rock
- Cracks (fissures) allow trapped HC's to flow into the well

U.S. SHALE PLAYS



U.S. Field Production of Crude Oil

Thousand Barrels per Day



eia

Data source: U.S. Energy Information Administration

U.S. 'Shale revolution'
(late 2000's)

UPSTREAM VS DOWNSTREAM

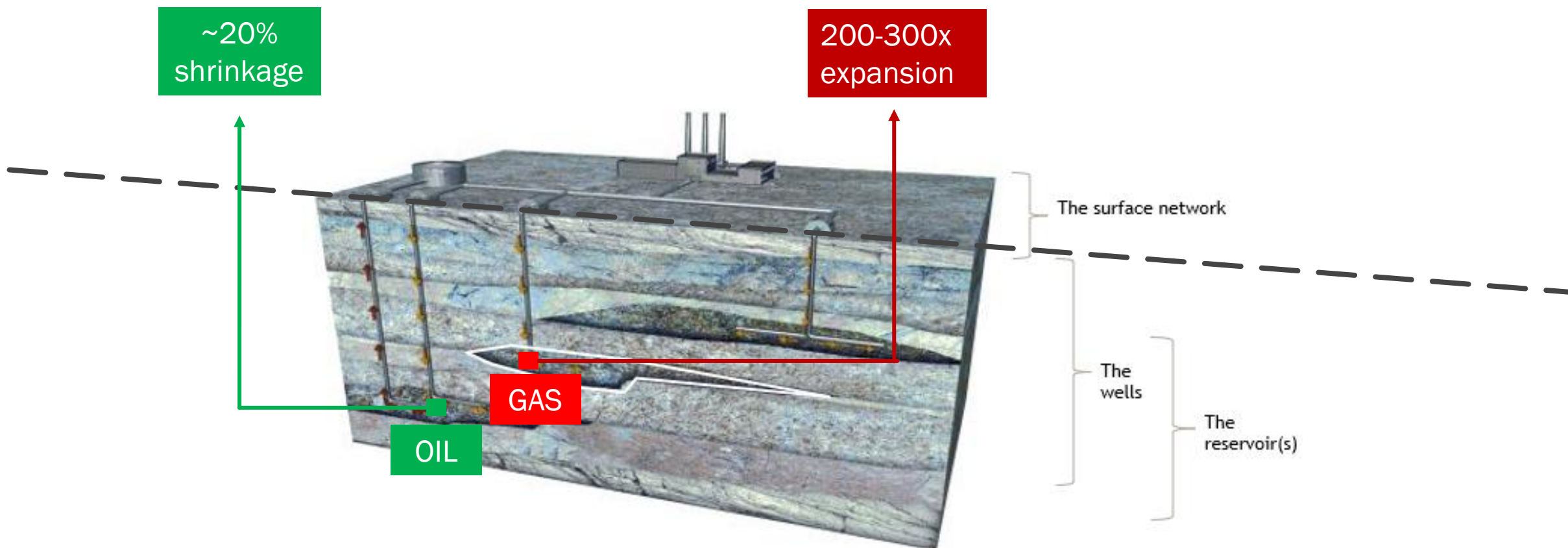


Exploration & Production
(E&P)

Storage & Transportation

Refining & Marketing

OIL VS GAS



CRUDE STORAGE

- Field-Level Storage (Upstream)
 - Temporary tanks near wells or gathering stations
- Terminal & Hub Storage
 - Large-scale tank farms
 - e.g., Cushing, Oklahoma

→ Crude is liquid:

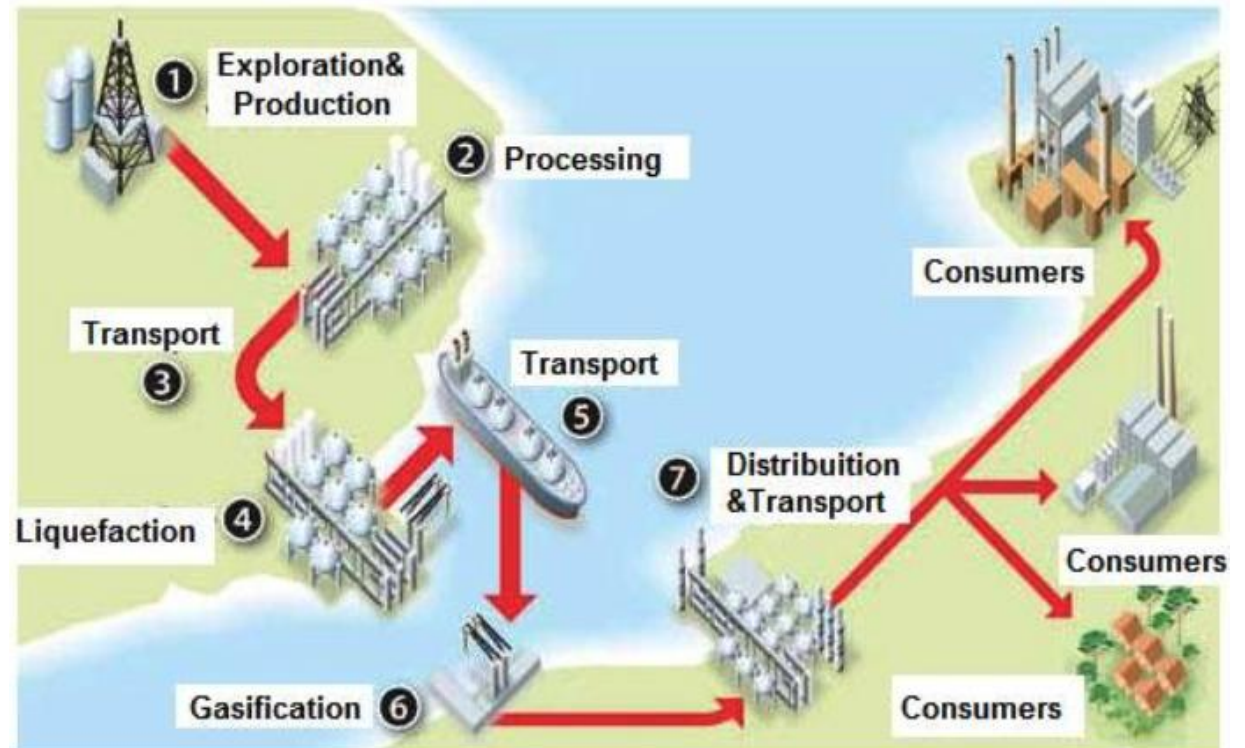
- Easily stored!



~90 million bbl storage capacity

GAS / LNG VALUE CHAIN

- It expands 200-300 times
 - Can't sit around
- Needs pressurization or cooling
 - LNG – cooled to -160 deg C to turn into liquid and ship in tankers
- Massive infrastructure needed
 - Pipelines
 - LNG terminals/ships/re-gas plants
- Needs market in advance
 - Sold before produced → Long-term contracts
 - Chicken-and-egg
 - No gas dev without a buyer
 - No buyer if no infrastructure



Gexcon AS

CRUDE MISMATCH– THE U.S. CONUNDRUM

- Crude types:
 - Light vs Heavy
 - Sweet vs Sour
 - U.S Refinery System built for Heavy Sour crude
 - Billions of \$ in upgrades to process heavy, sour crude
 - Imports from Venezuela, Mexico, Canada
 - Shale revolution in the U.S.
 - The problem: Shale Oil → Light and sweet
 - Crude mismatch: incompatible with U.S. refineries
 - Shale Oil is not substitute!
 - US needs Canada and Venezuela!
- 
 - Geopolitical implications
 - Venezuela & Canada are strategic suppliers
 - Venezuela
 - Under U.S. sanctions
 - Exports to China
 - Canada
 - >60% U.S. imports
 - Maxed out
 - Trump's tariffs
 - Leverage

PERU-CHILE LNG TRADE PARADOX

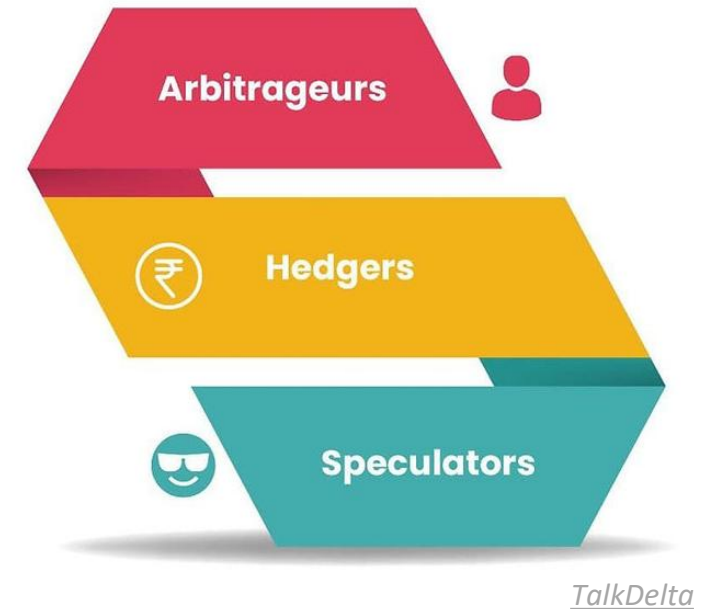
- Peru exports LNG to Europe, bypassing neighbor Chile
- Chile imports LNG from the U.S., sailing around the continent
- No infrastructure or LNG swap exists between them
- Long-term contracts + weak regional integration block direct trade
- Why?
 - Lack of regional energy coordination
 - Limited political leverage
 - Ineffective geopolitical framework for resource sharing



@anasalhajji

HOW IS OIL PRICED?

- Oil is priced in the Futures market
- What is a futures contract?
 - An agreement to **buy or sell something** (like oil) at a **set price** on a **specific future date**.
 - **Producers** (like oil companies) protect themselves from falling prices
 - **Buyers** (like refineries) avoid sudden price spikes
 - **Traders** and **speculators** bet on future price movements to make profit
 - Traded in the Open market
 - NYMEX
 - ICE (London)
- Benchmark prices
 - Brent, WTI, Dubai/Oman
 - Price: benchmark + adjustments (quality differential + transport cost)



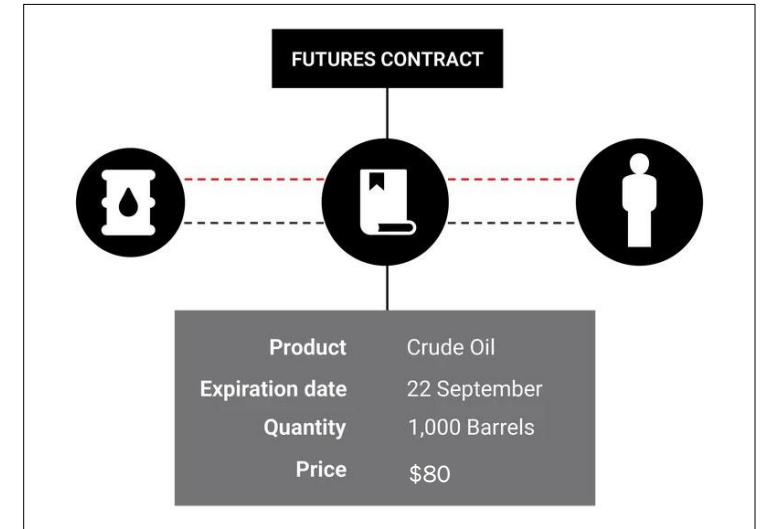
EX: HEDGING

- You're an oil producer.
- In June (today), you see that the September futures is trading at \$80/barrel
- You worry that prices might drop to \$70 by then
 - You'd like to secure the \$80/b price

The Hedge

- You sell today a futures contract at \$80/b
- If in Sep price did drop to \$70:
 - You sell your oil in spot market at \$70
 - You close down the futures contract buying it back at \$70 (current price)
 - You had sold it at \$80, so make \$10 on the future
- If price went the other way, say, to \$90
 - Money is lost on the hedge
 - It's the cost of 'insurance'

} Cash-settled



} Locking in a price today for future delivery

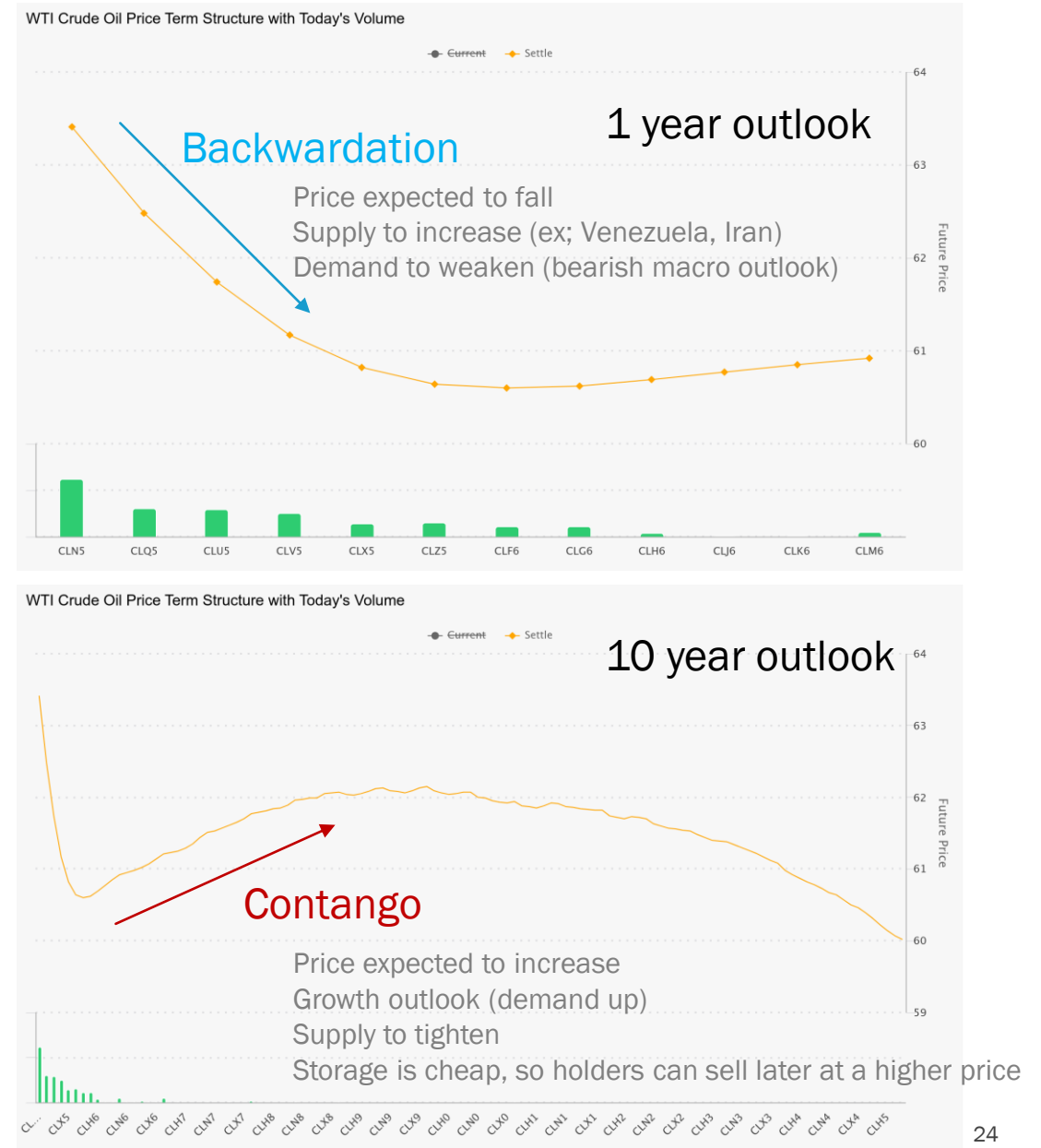
CRUDE OIL PRICE



- Front month (~spot)
- Brent > WTI ~\$2-6/b
 - Logistics & global markets

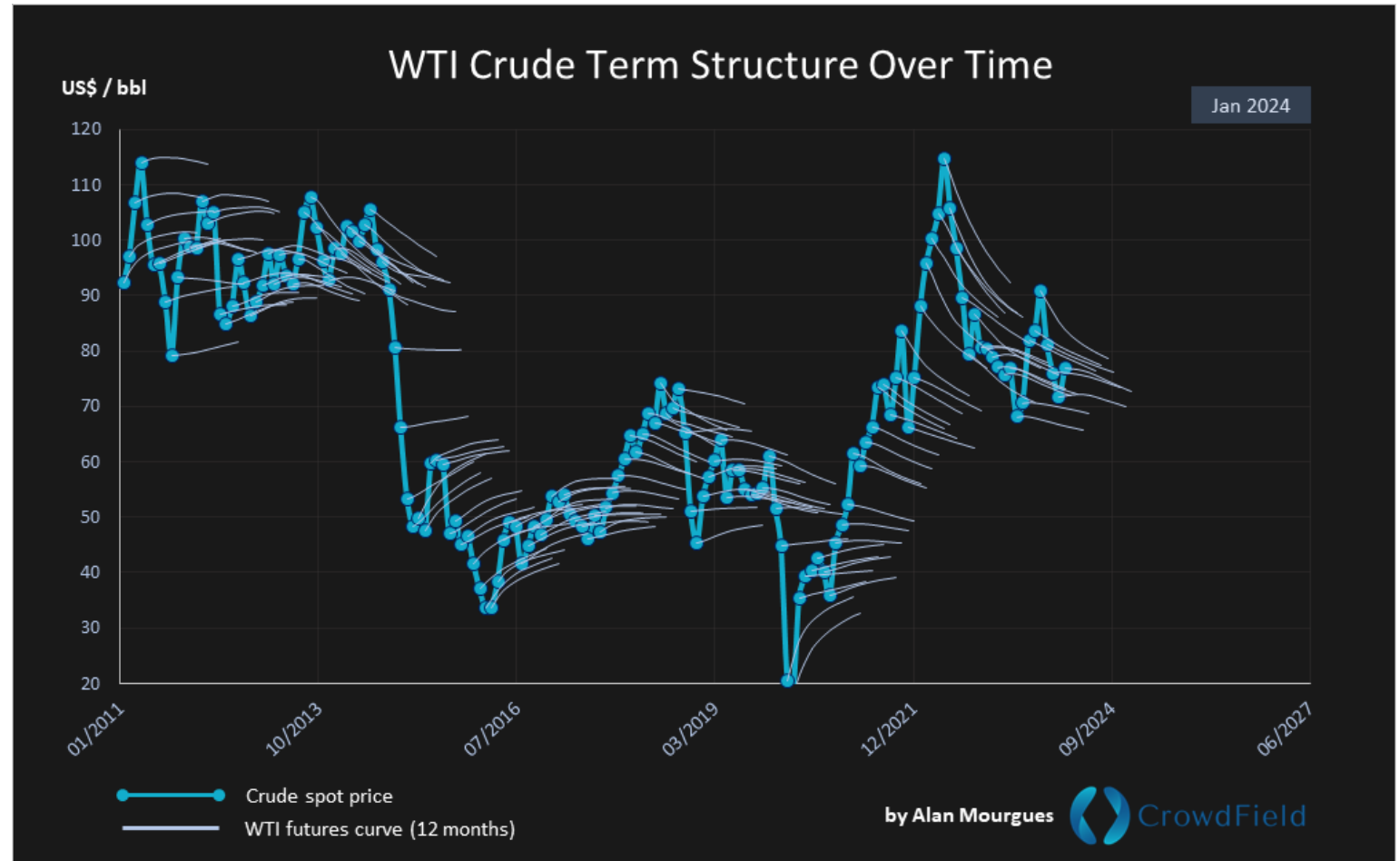
TERM STRUCTURE

- Curve with the 'future price of oil'
- One value per month, looking 10 years into the future
- Technically speaking, it's not where the spot price will be in the future, but rather the current price for future delivery.
 - Since spot converges toward the future price at expiration, it can be thought of as the future price of oil



PRICE PREDICTION

- A fools' errand
- Nobody can predict the oil price
- The market gets it wrong all the time!



OIL PRICE DRIVERS

Market Fundamentals

- Supply shocks: Wars, sanctions, weather events (e.g., hurricanes)
- Demand swings: Economic growth, recessions, fuel seasonality

Producer Influence

- OPEC+ policy decisions: Output targets, quota adjustments
- Production outages: Maintenance, strikes, underinvestment

Geopolitical & Trade Factors

- Conflicts, sanctions, and political unrest
- Transportation constraints: Blocked canals, congested ports

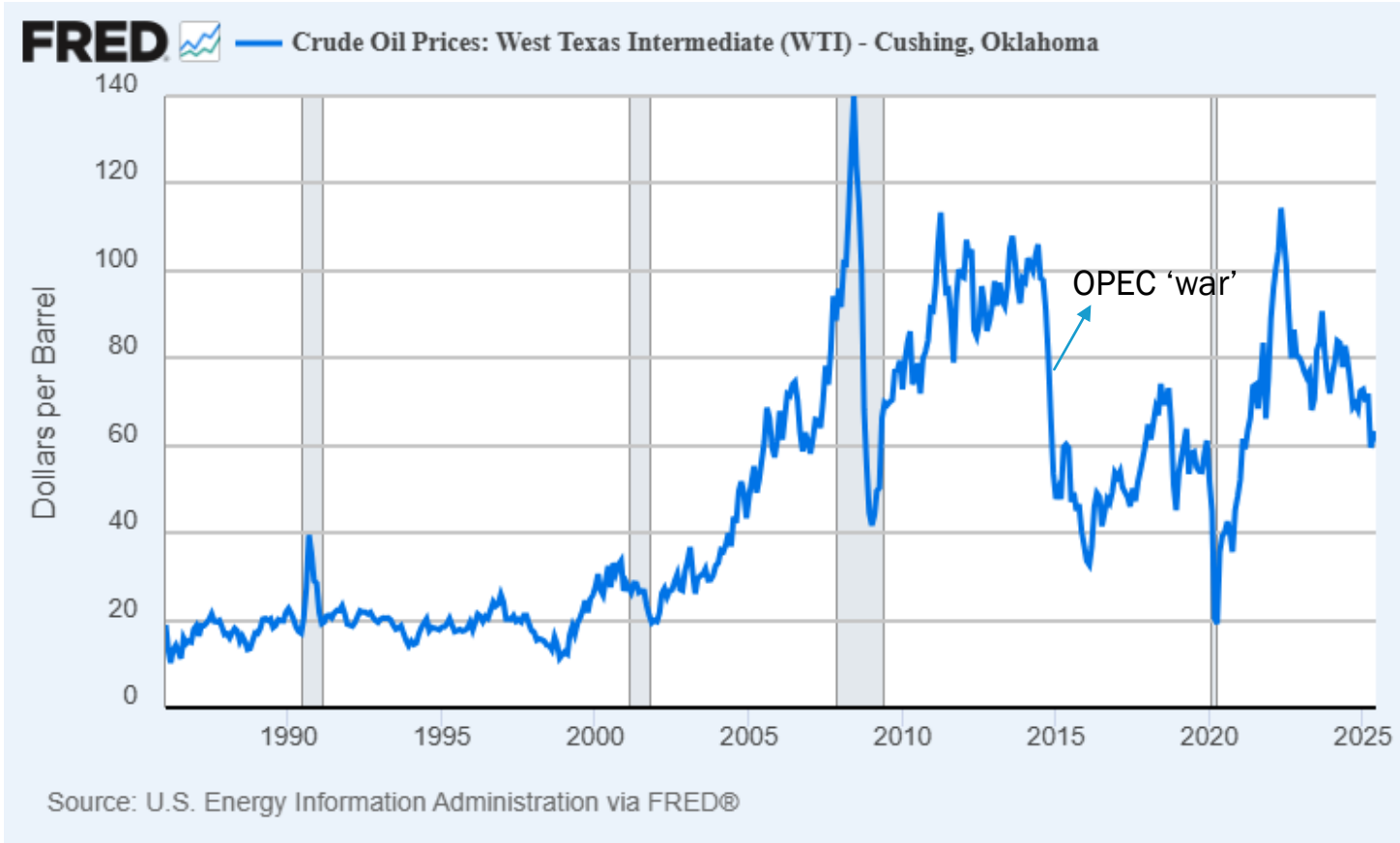
Market Conditions

- Storage levels & inventories
- Macroeconomic variables:
 - Currency strength (e.g., USD value)
 - Interest rates
 - Inflation expectations
 - Policy and regulatory uncertainty

Financial Markets

- Futures trading and speculative flows
- Algorithmic and hedge fund activity

CRUDE AND RECESSIONS



- Recessions (gray bars) typically align with sharp drops in oil prices
- Oil prices are highly cyclical peaking during economic booms and crashing during demand shocks

2014–2016 Oil Price Crash

- In late 2014, OPEC chose not to cut production
- Oversupply of oil
- Why? → Defend market share
- Response to US shale oil boom

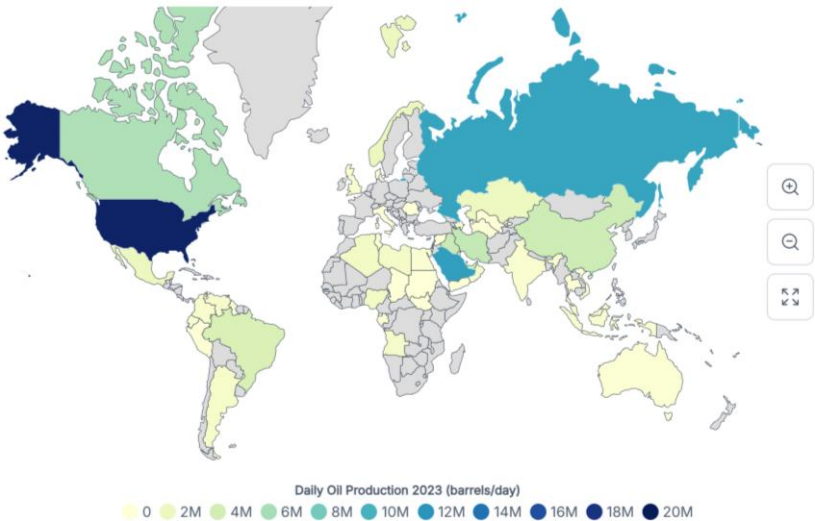
COVID & UKRAINE



OIL PRODUCTION – TOP 10 COUNTRIES

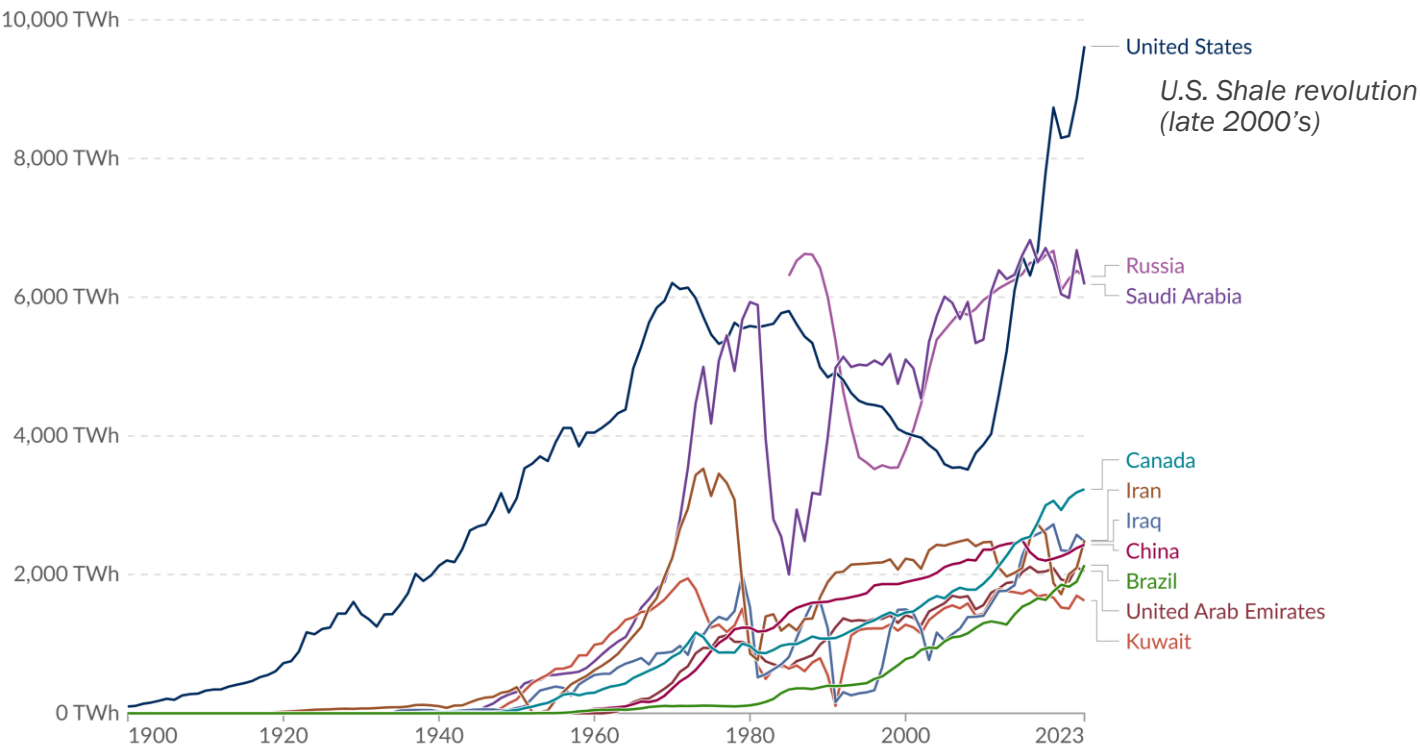
COUNTRY	DAILY OIL PRODUCTION 2023 (BARRELS/DAY) ↓
 United States	19.4M
 Saudi Arabia	11.4M
 Russia	11.1M
 Canada	5.7M
 Iran	4.7M
 Iraq	4.4M
 China	4.2M
 United Arab Emirates	3.9M
 Brazil	3.5M
 Kuwait	2.9M

Source: worldpopulationreview.com/



Oil production

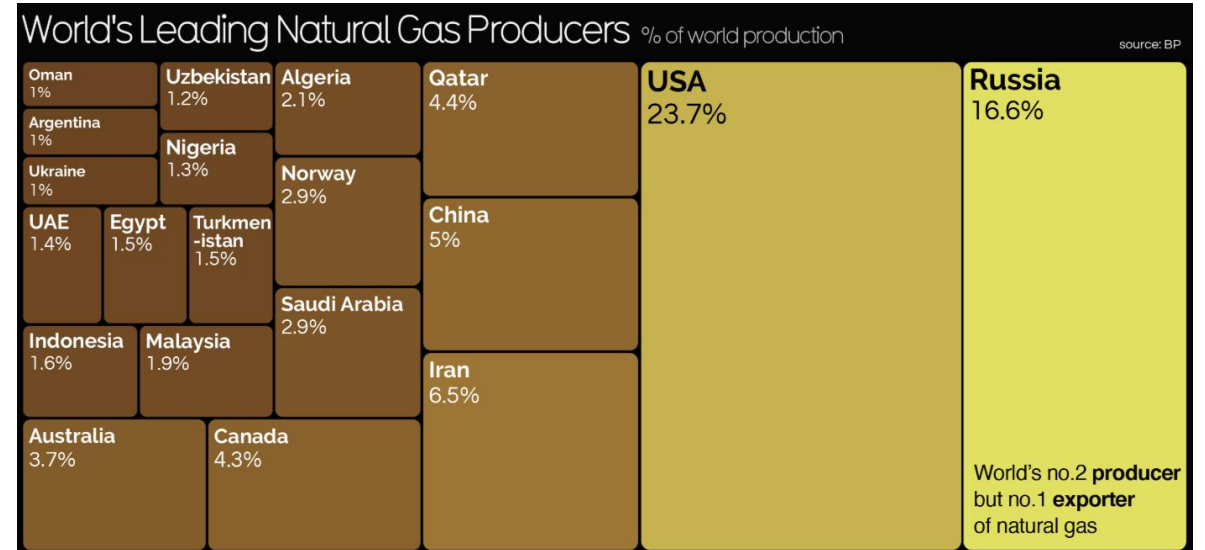
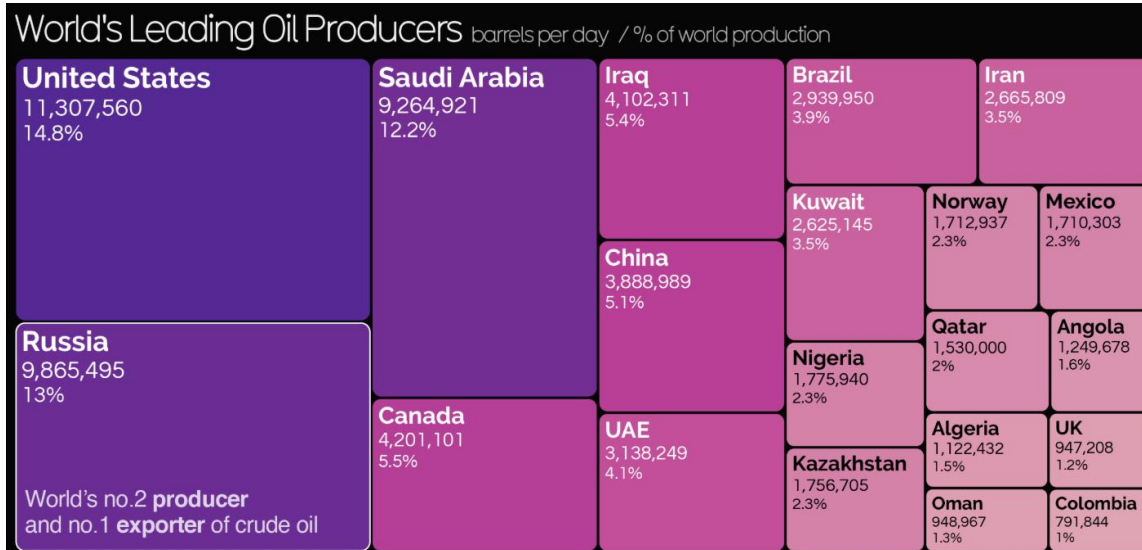
Measured in terawatt-hours.



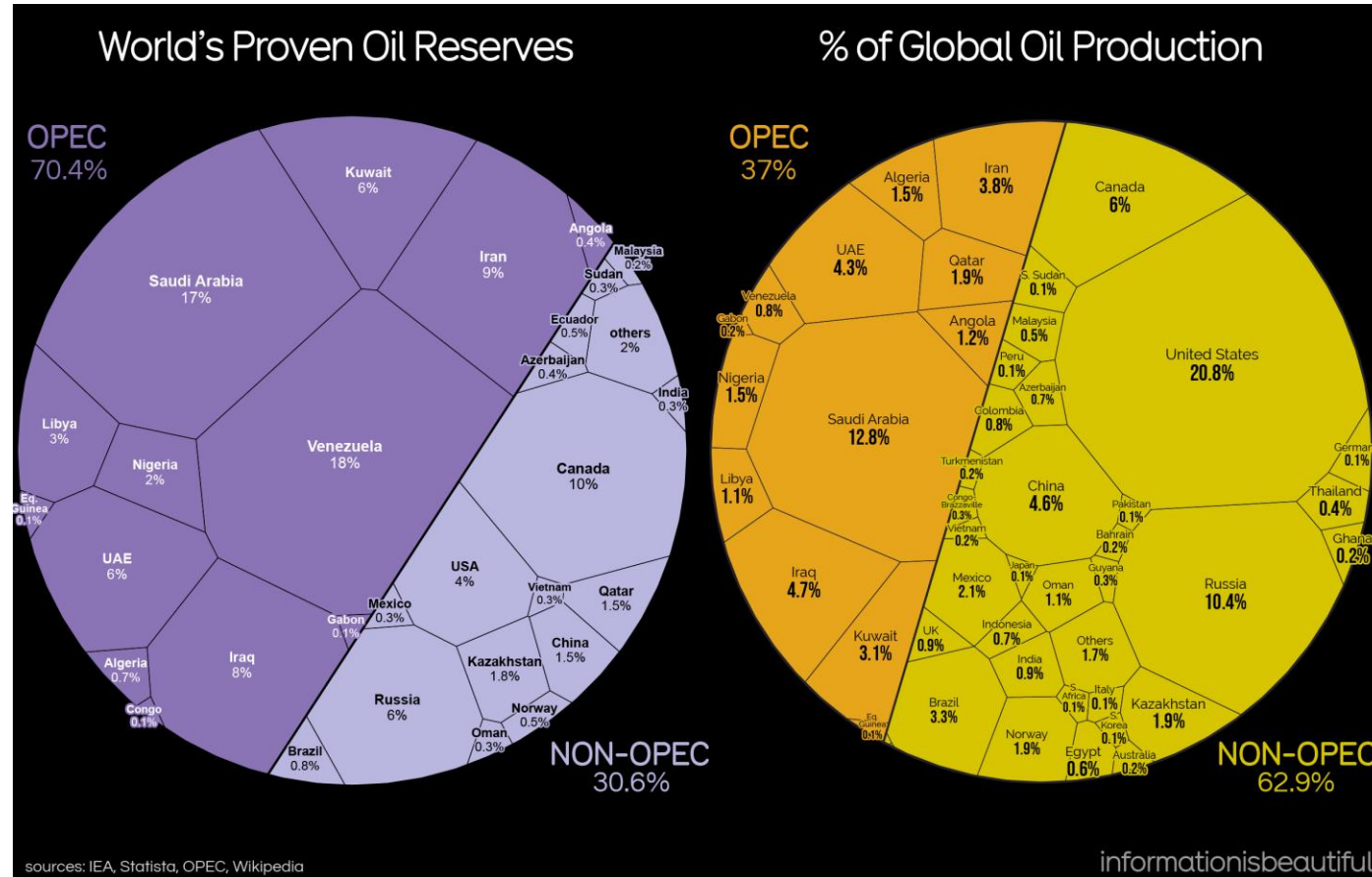
Data source: Energy Institute - Statistical Review of World Energy (2024); The Shift Data Portal (2019)

OurWorldinData.org/fossil-fuels | CC BY

WORLD'S OIL & GAS PRODUCERS

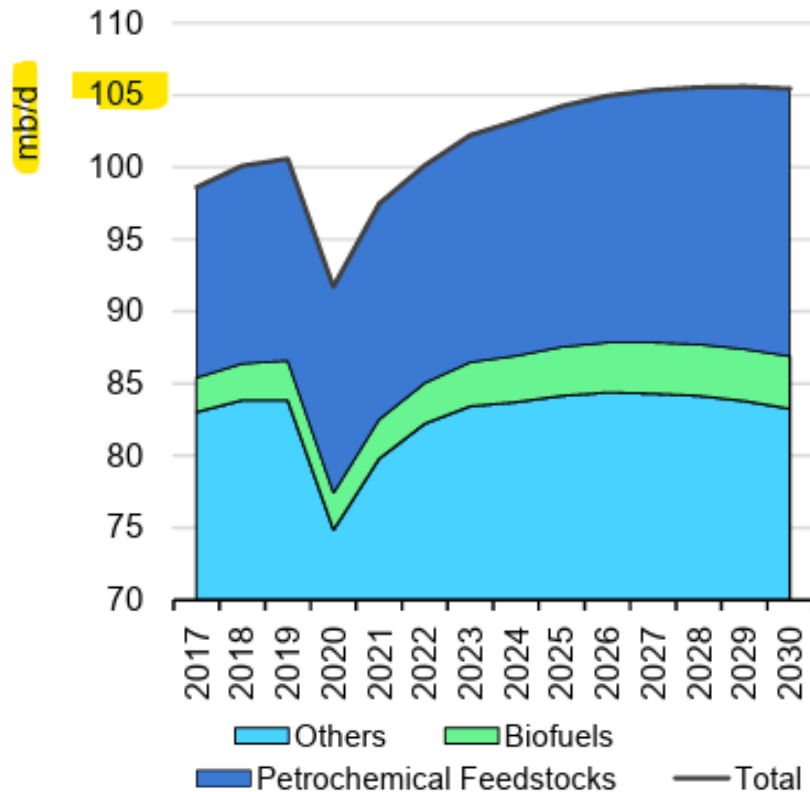


PRODUCTION & RESERVES



- OPEC owns 70% of proven reserves, but produces only 37% of global oil
- The U.S. has just 4% of reserves, yet is the top producer (21%) — thanks to shale
- Venezuela has the most reserves but produces very little (0.8%) due to underinvestment and sanctions

OIL DEMAND FORECAST



IEA

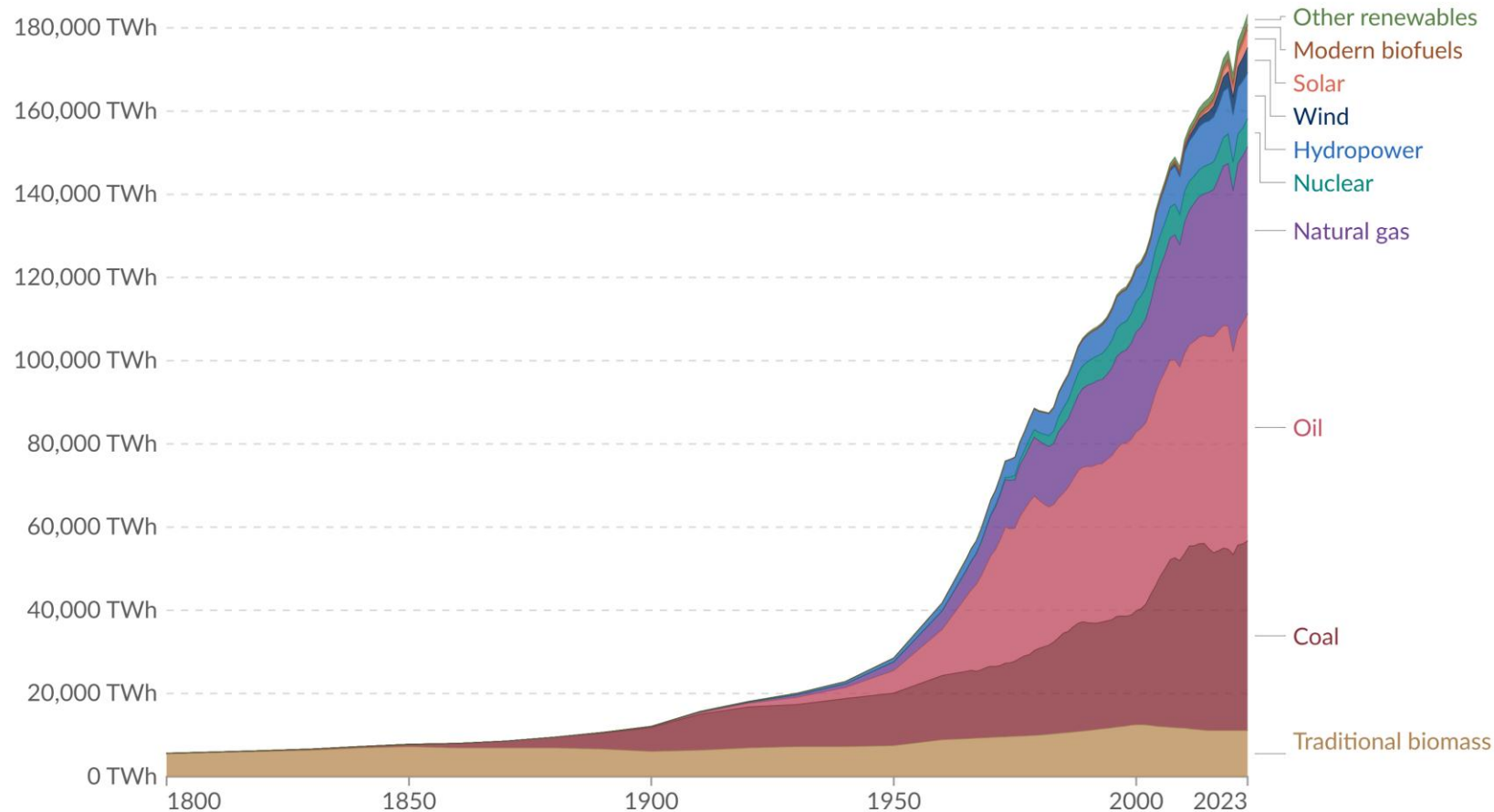
- Oil demand is expected to level off near 106 mb/d by 2030, due to the clean energy transition.
- EV growth and fuel efficiency are cutting oil use in transport.
- Oil's role in power generation is shrinking, replaced by renewables and gas.
- Demand will still rise in jet fuel and petrochemicals, especially LPG, ethane, and naphtha

ENERGY MIX

Global primary energy consumption by source

Our World
in Data

Primary energy¹ is based on the substitution method² and measured in terawatt-hours³.

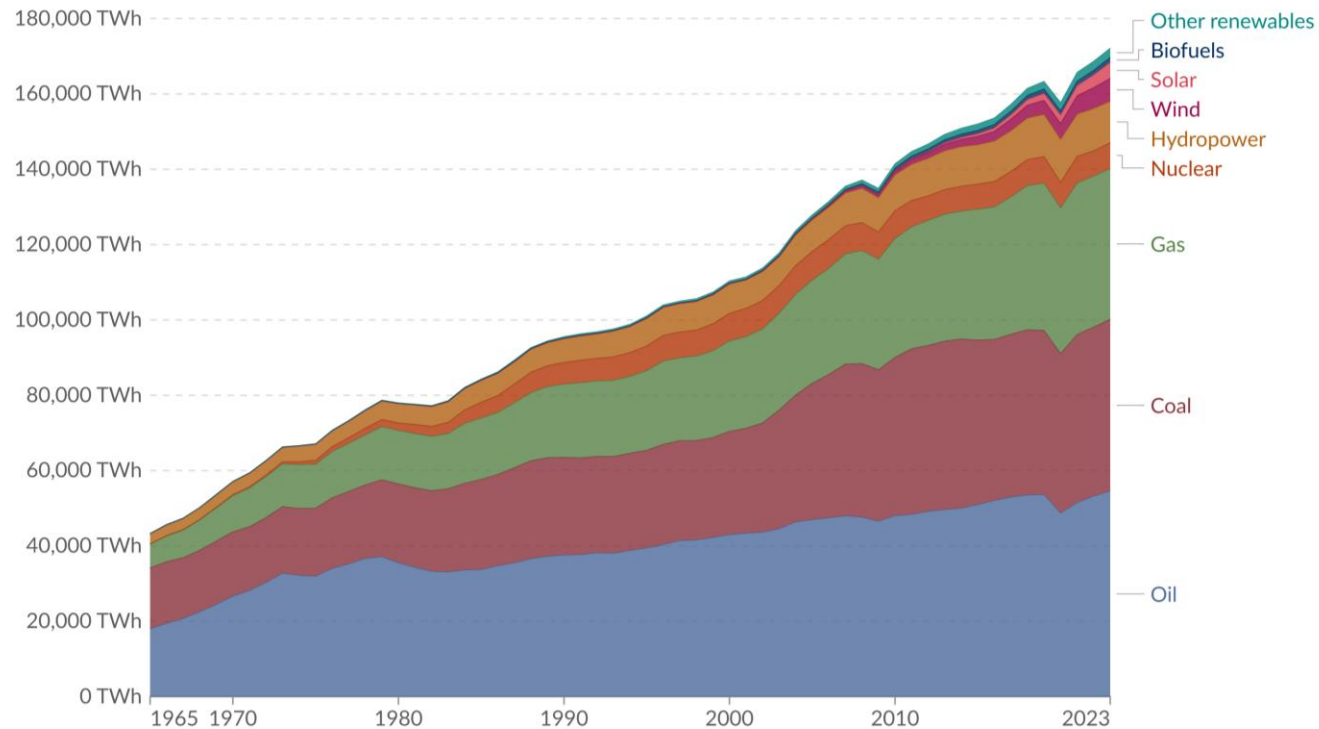


Data source: Energy Institute - Statistical Review of World Energy (2024); Smil (2017)

OurWorldinData.org/energy | CC BY

Note: In the absence of more recent data, traditional biomass is assumed constant since 2015.

ENERGY MIX



Data source: Energy Institute - Statistical Review of World Energy (2024)

Note: "Other renewables" include geothermal, biomass, and waste energy.

OurWorldinData.org/energy | CC BY

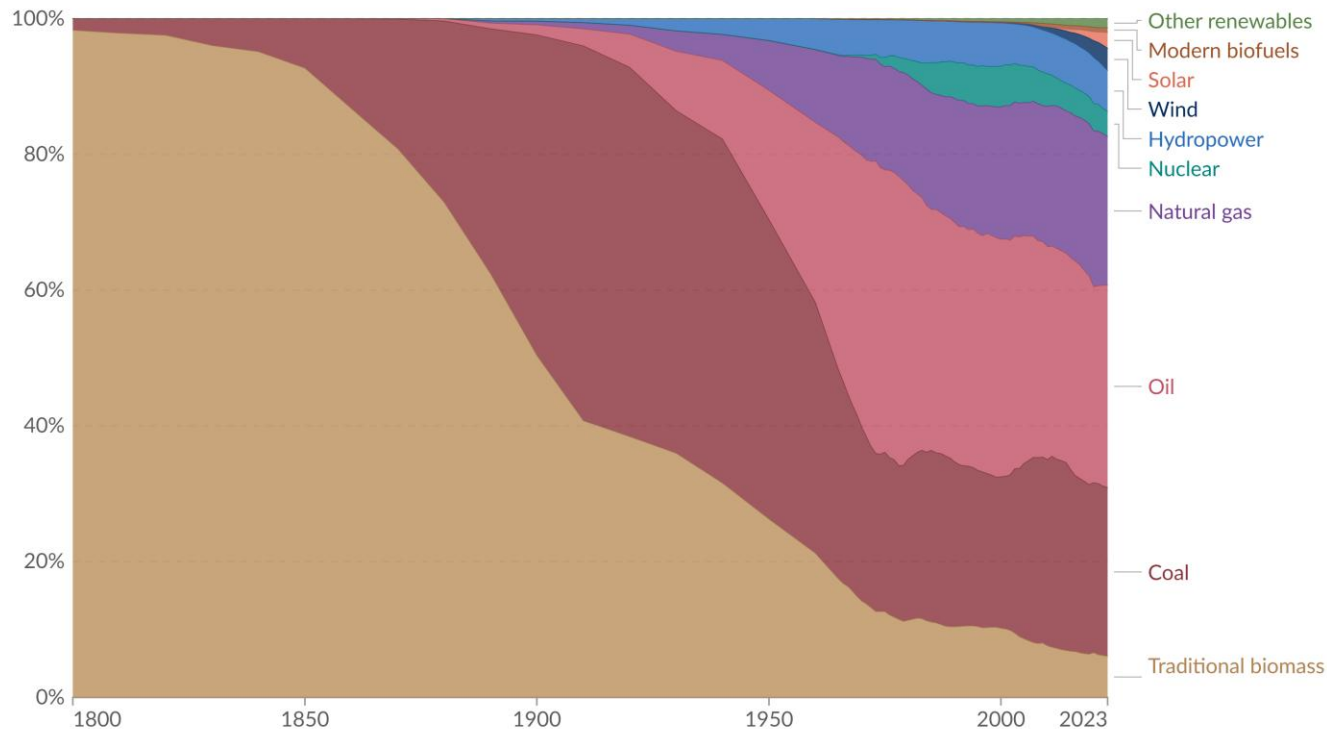
- Energy demand has quadrupled since 1965.
- Oil, coal, and gas still dominate the mix.
- Gas use has steadily grown, now close to coal.
- Solar and wind are rising, but still small overall.
- Fossil fuel use is still growing in absolute terms.

ENERGY SUBSTITUTION

Global primary energy consumption by source

Primary energy¹ is based on the substitution method² and measured in terawatt-hours³.

Our World
in Data



Data source: Energy Institute - Statistical Review of World Energy (2024); Smil (2017)

Note: In the absence of more recent data, traditional biomass is assumed constant since 2015.

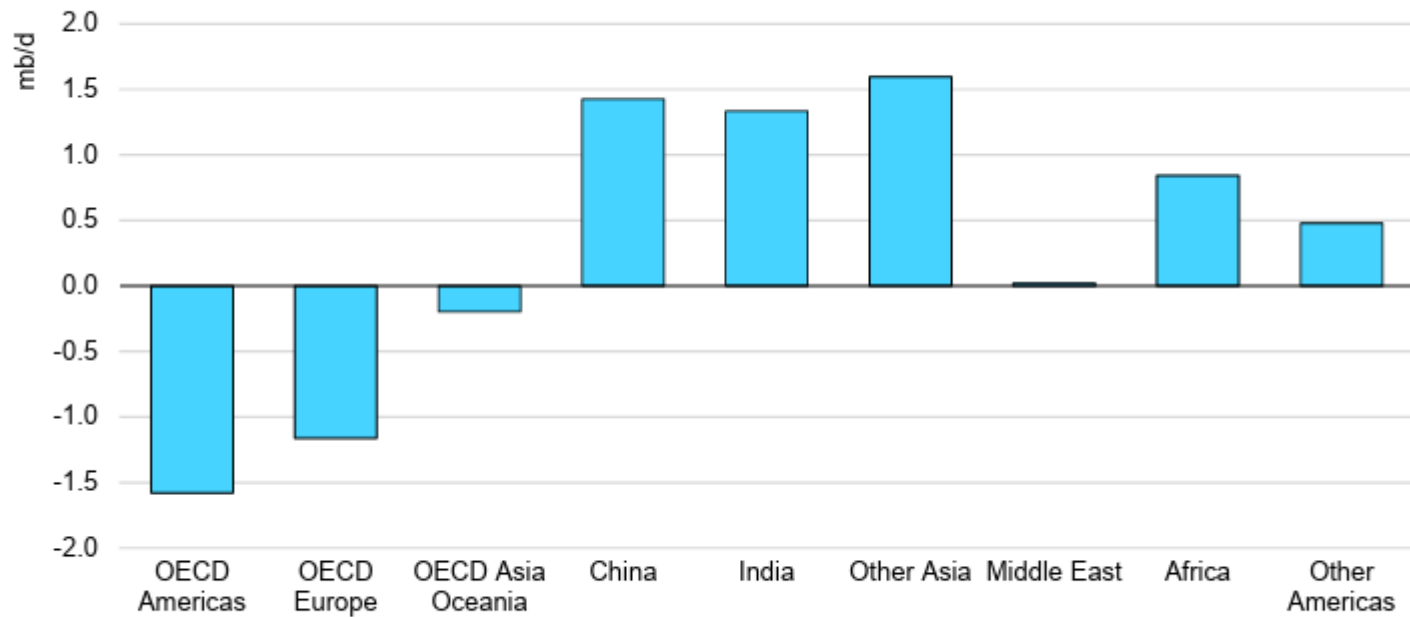
OurWorldinData.org/energy | CC BY

- Long ago, we mostly burned wood and plants for energy.
- Then came coal, and it took over fast.
- Oil showed up and became king by the mid-1900s.
- Later, gas, hydro, and nuclear joined the mix.
- Now we're adding wind and solar, but they're still small today.

→ Each new energy source takes market share from the prior ones, making the prior ones look like they're getting phased out.. But they stick around

OIL DEMAND PER REGION

Growth in oil demand by region, 2023-2030

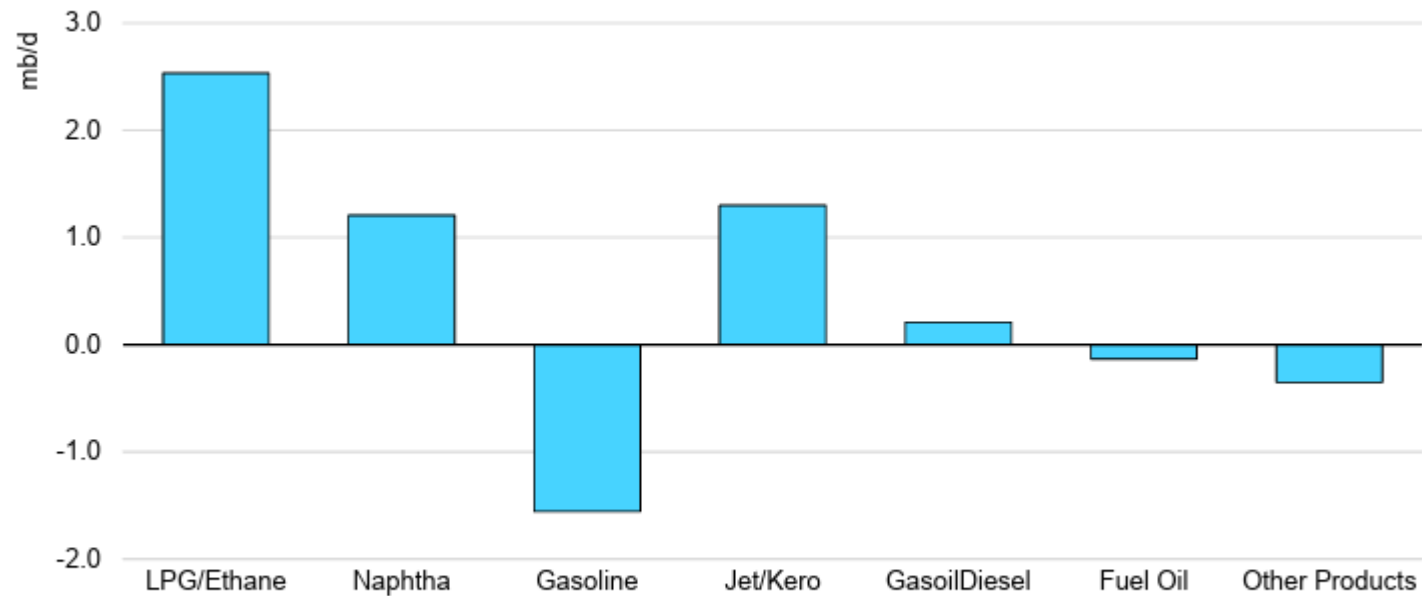


IEA. CC BY 4.0.

- Oil demand is moving away from wealthy countries and growing in emerging markets.

OIL DEMAND PER PRODUCT

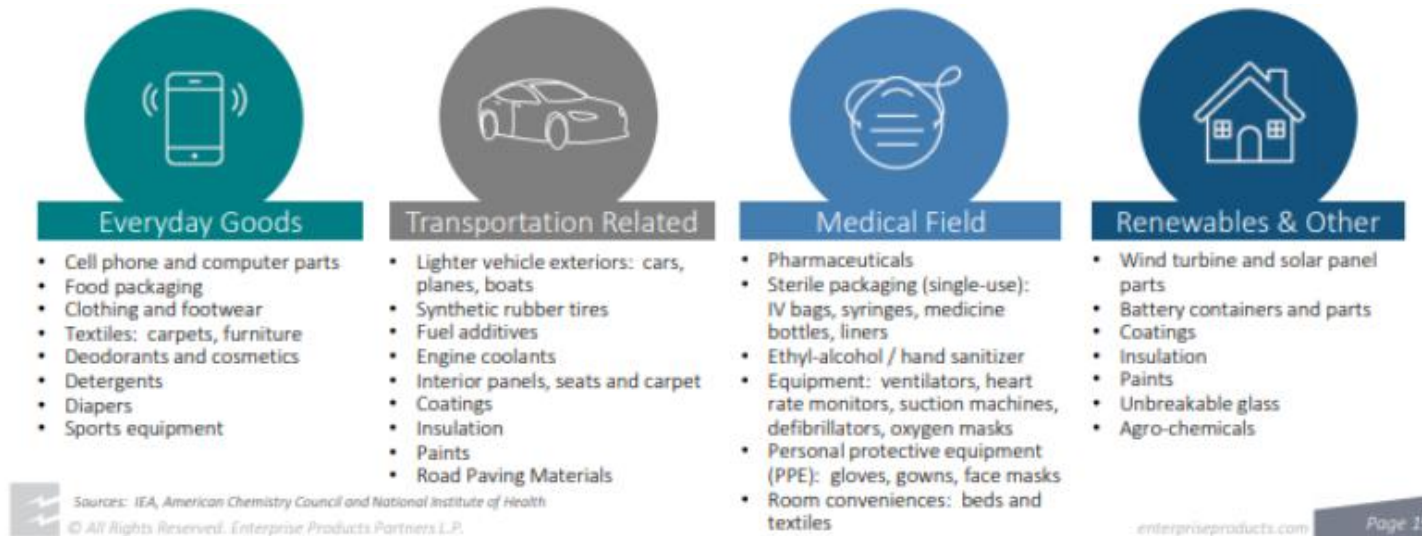
Growth in world oil demand by product, 2023-2030



IEA. CC BY 4.0.

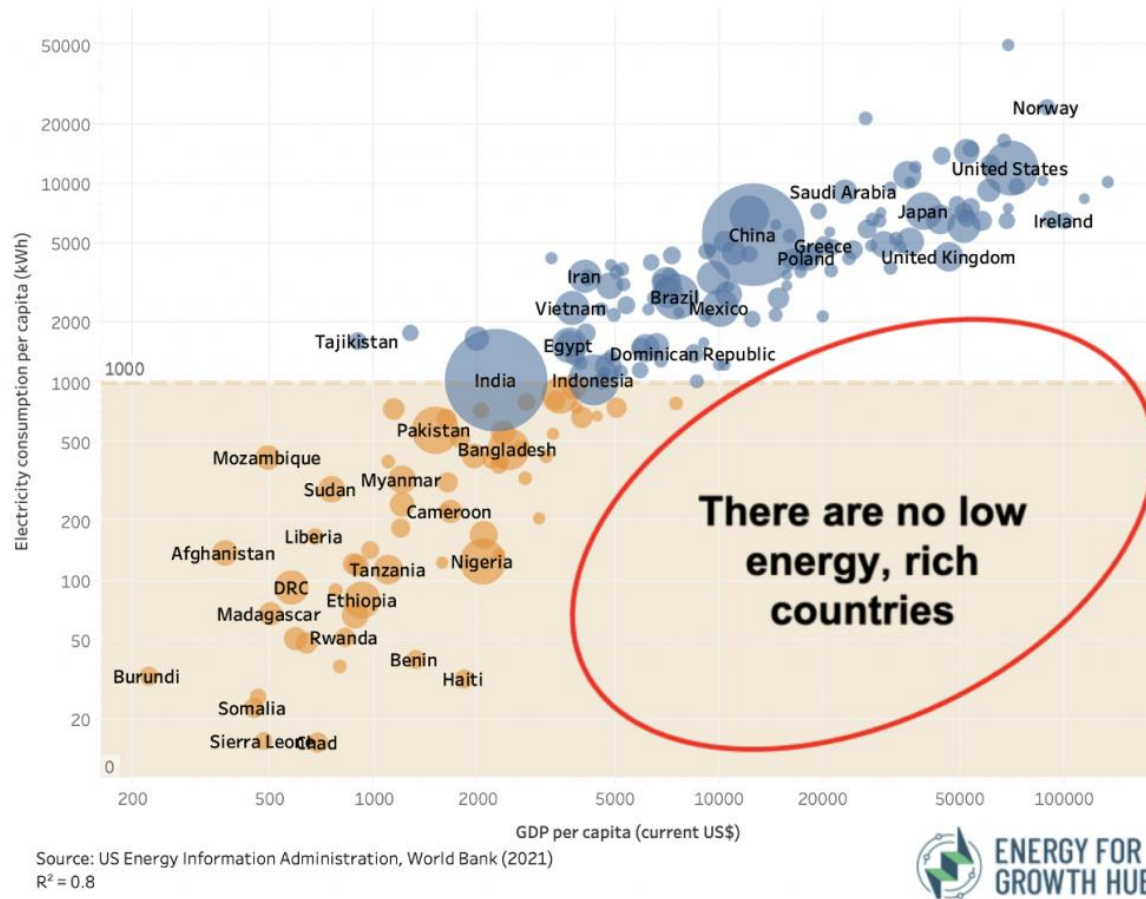
- Oil demand shifts away from fuels for vehicles and towards feedstocks for chemicals and plastics.

PETROCHEMICALS



- Petrochemicals will drive 60% of oil demand growth over the next decade
- 96% of manufactured goods rely on petrochemicals
- 99% of pharmaceuticals use petrochemical inputs
- Plastics make up 50% of car volume, but only 10% of the weight
- Just 2 lbs of plastic can hold 10 gallons — far lighter than glass, steel, or aluminum

ENERGY VS GDP



- Nowhere in the world is there a wealthy country that consumes only a little energy, nor a poor country that consumes a lot.

OIL IS HARD TO REPLACE

- Growing economies need cheap, reliable energy — oil still delivers both
- Most transport still runs on oil, especially aviation, shipping, and heavy freight
- Oil is not just fuel — it's a feedstock for plastics, fertilizers, and chemicals
- Petrochemical demand is rising, even as fuel demand plateaus



THE EV ILLUSION

The Hidden Oil in Electric Vehicles



- EVs cut gasoline use, but...
 - EVs are built with petrochemicals:
 - Plastics, composites, synthetic rubbers, adhesives, sealants
 - Lightweight materials increase petrochemical demand
 - Charging stations: made with cables, plastics, and resins — all oil-derived
 - Electricity generation is still heavily fossil-fueled in much of the world
- Bottom line: Oil demand doesn't disappear — it shifts from the fuel tank to the factory and grid

IF NOT OIL & GAS, THEN WHAT?

■ Biomass:

- Low energy return
- Environmentally harmful (fertilizers, land use, water pollution)

■ Coal

- Air pollution and health issues
- High CO2 emissions

■ Wind

- Intermittent and low power density
- Can't provide baseload power
- Not recyclable. Blades end in landfills

■ Solar

- Long payback times, Low energy return
- Not recyclable. Toxic materials from old panels

■ Hydro

- Geographically limited
- Ecological disruption

■ Geothermal

- Location-specific
- Costly and underdeveloped

■ Nuclear

- Rare but high-impact accidents
- Long-lived waste disposal
- Underfunded in innovation

TRANSITION

- Renewables are growing fast, but fossil fuels still supply over 80% of global energy
- Developing countries need affordable and reliable energy to grow — often still fossil-based
- Energy infrastructure (refineries, grids, power plants) is built to last 30–50 years
- The global energy system is massive — replacing it takes decades, not years
- The transition isn't just technical — it involves policy, geopolitics, economics, and behavior



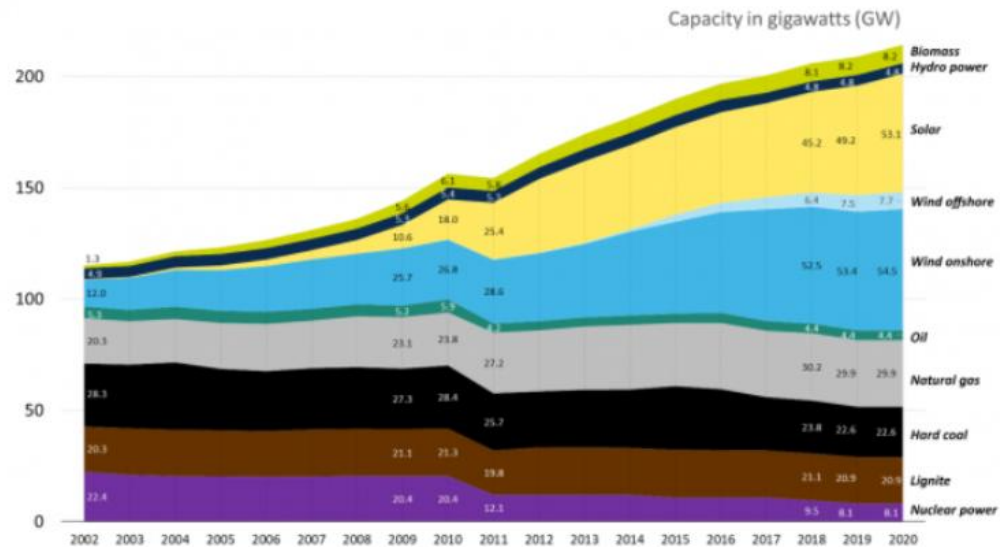
OurFuture.Energy

CASE STUDY: GERMANY

- Germany has been at the forefront of shifting to wind & solar for electricity generation.

Installed net power generation capacity in Germany 2002 - 2020.

Data: Fraunhofer ISE 2020.



- However, electricity is only ~20% of total energy consumption.
 - Still reliant on fossil fuels for all other energy uses
- Despite all this effort, coal usage barely dropped, and natural gas usage increased
 - Total fossil reliance has stayed mostly flat.
- They also phased out their Nuclear capacity, post-Fukushima
 - Emphasizing wind and solar for their grid.

CASE STUDY: GERMANY

Actual results

- Germany has among the highest electricity costs in the world.
- Energy surplus built over two decades vanished
 - In 2023, Germany became a net electricity importer for the first time in two decades
- Nuclear energy was replaced by coal and imports. The social cost is estimated at \$12B/year, mostly from higher mortality due to air pollution (NBER 2019 study).
- Solar: ethical supply chain issues (China).
- Wind and solar contribute to e-waste and landfills.
- As heating and transport electrify, grid demand will surge—a big challenge if relying solely on intermittent renewables.

CASE STUDY: INDIA

- Fast growth = additive, not replacement
 - In rapidly growing economies like India, new energy sources (like solar) are added rather than replacing old ones (like coal).
- Solar booming, but coal booming faster
 - Despite explosive solar capacity growth, coal has grown even more in absolute electricity generation (TWh).

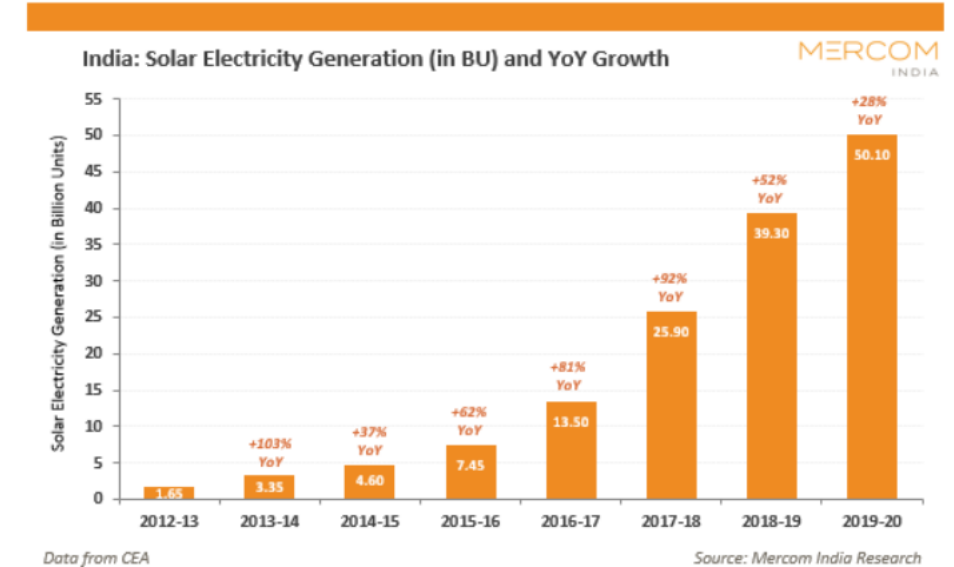
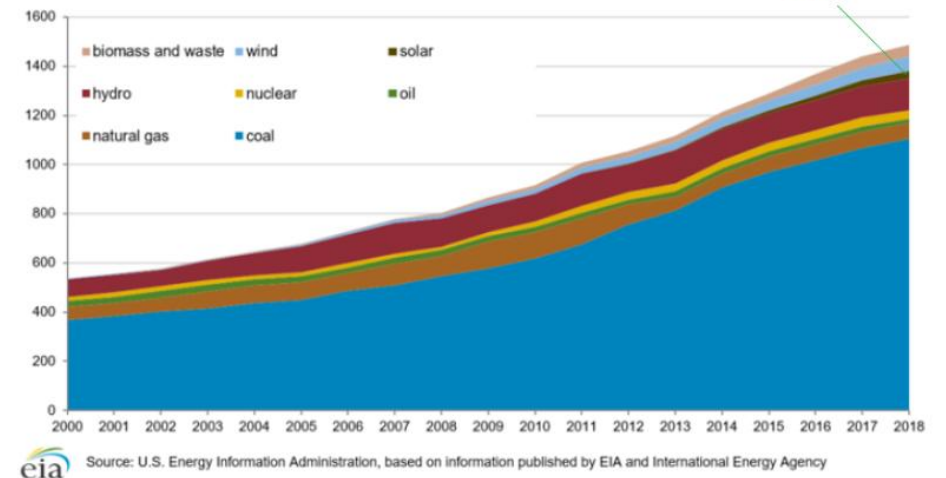
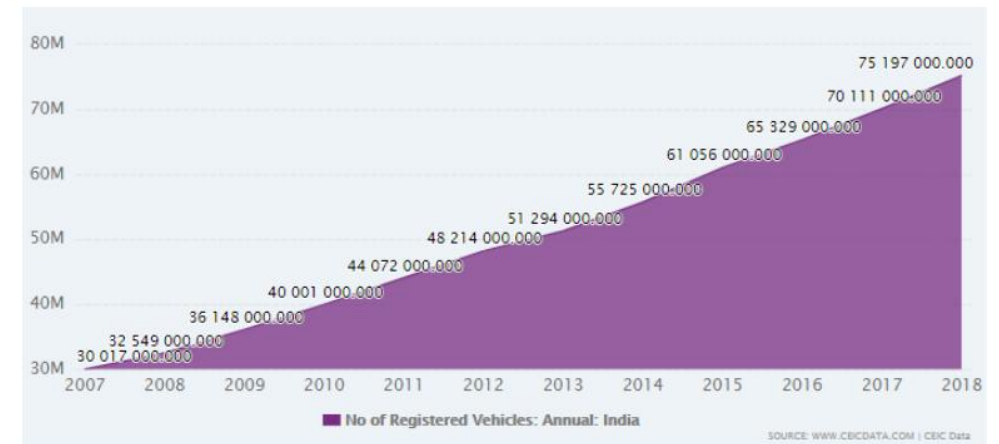


Figure 8. India's net electricity generation by fuel type, 2000-2018

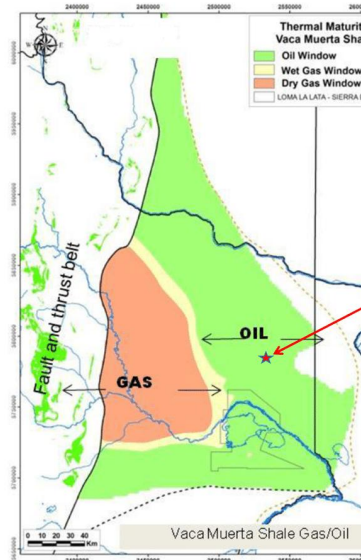


CASE STUDY: INDIA

- Why coal remains dominant:
 - Fast ROI and payback
 - High power density
 - Anywhere, regardless of climate
 - Domestic coal reserves
- Coal's downside:
 - High pollution and CO₂ emissions, with severe environmental and public health impacts.
- Grid ≠ total energy:
 - Electricity is only part of India's energy picture. Hydrocarbons (LPG, gasoline, diesel) are still expanding for cooking and transport.
- Cars on the rise:
 - Passenger vehicle numbers are growing fast, mostly gasoline-powered.
 - Even EVs would still mostly run on coal-fired electricity today



ARGENTINA – VACA MUERTA



TOP 5 SHALE GAS

Country	Trillion cubic feet	Global Share
China	1115.2	14.7%
Argentina	801.5	10.6%
Algeria	706.9	9.3%
U.S.	622.5	8.2%
Canada	572.9	7.6%

- World's 2nd largest Shale Gas resource

TOP 5 SHALE OIL

Country	Billion barrels	Global Share
U.S.	78.2	18.7%
Russia	74.6	17.8%
China	32.2	7.7%
Argentina	27.0	6.4%
Libya	26.1	6.2%

- World's 4th largest Shale Oil resource

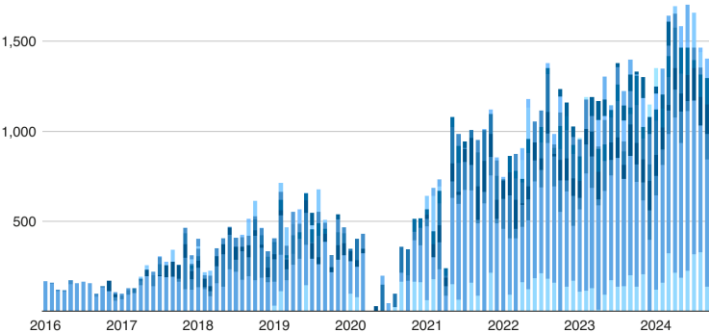
Source: Forbes ('technically recoverable')

VACA MUERTA BOOM

Argentina: Fracking Boom

The monthly number of fracking wells for non-conventional oil and gas in Argentina has boomed in recent years, topping over 1,500 in 2024 as energy firms have flocked to the Vaca Muerta formation.

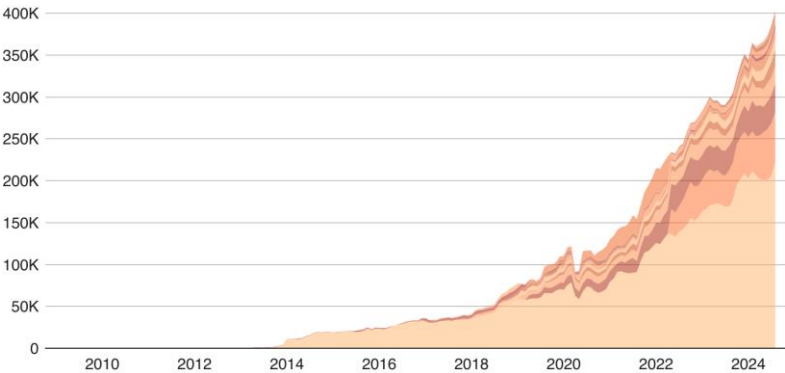
● VISTA ● YPF ● Tecpetrol ● PAE ● Shell ● Pluspetrol ● Total ● ExxonMobil ● Phoenix ● Pampa Energia
 ● Chevron ● CAPEX ● Wintershall ● CGC



Note: Number of fracking wells per month
 • Source: Aleph Energy

Argentina: Shale Oil Production

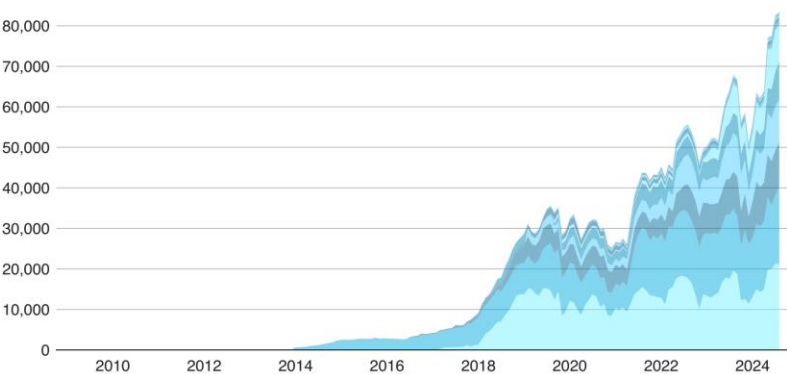
Argentina's shale oil production has climbed sharply in recent years. Now a new libertarian President Javier Milei is rolling out incentives and encouraging firms to drill more wells.



Note: Data is barrels per day
 • Source: Aleph Energy

Argentina: Gassing Up

Argentina's shale gas production has shot up this year, signaling that the huge "Dead Cow" formation may finally be on track to fulfill its potential.

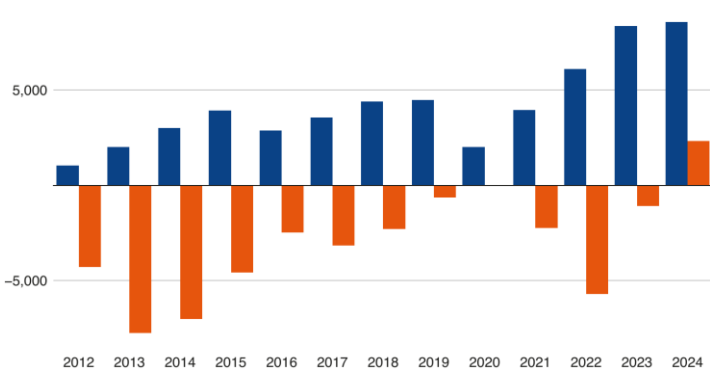


Note: Data is in thousands of cubic meters of gas production per day
 • Source: Aleph Energy

Argentina: Shale Investment and Energy Surplus

Argentina has seen investment in shale pick-up, helping boost production and putting the country on track for a rare energy trade surplus this year.

● Investment Shale Oil and Gas ● Energy trade balance



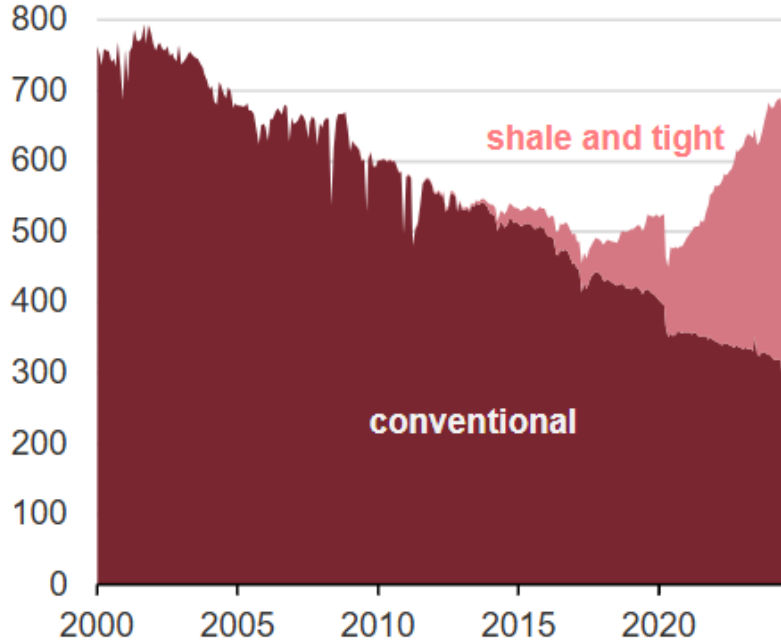
Note: Data is in millions of U.S. dollars
 • Source: Aleph Energy

UNCONVENTIONAL 'REVIVAL'

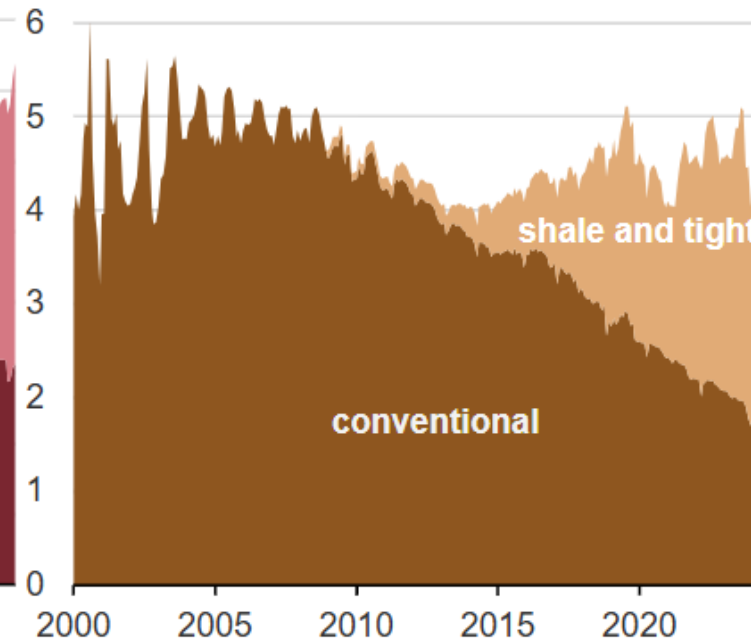
Argentina crude oil and natural gas production (Jan 2000–Sep 2024)



crude oil production
thousand barrels per day

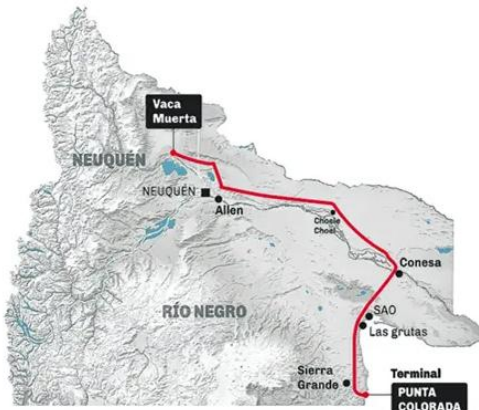


natural gas production
billion cubic feet per day

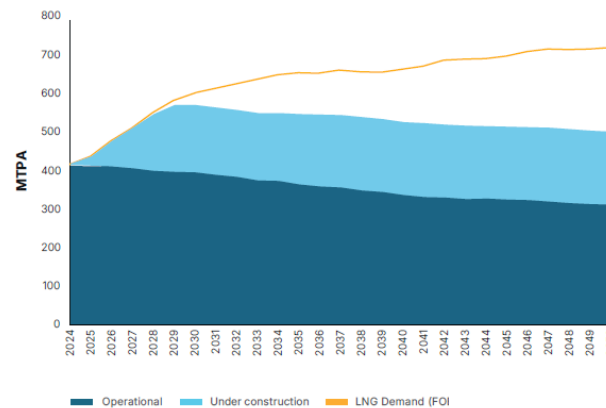


Data source: U.S. Energy Information Administration and Argentina Ministry of Energy (Secretaría de Energía de la República Argentina [SESCO])

UNLOCKING VACA MUERTA'S GAS POTENTIAL – TOUGH!



LNG supply-demand gap MTPA



LNG maritime routes



Source: YPF

- Massive Infrastructure investment (~\$60b)
 - Pipeline
 - Processing plants
 - LNG terminals
 - Multiple partnerships, negotiations, long-term contracts, etc
- Political instability: past contract breaches
 - Investors hesitant
- Falling gas prices: Global LNG prices are down sharply from 2022
 - long-term projects less attractive.
- Competing global LNG export capacity
 - LNG export capacity coming online in early 2030s from Qatar and the U.S.,

VACA MUERTA - SUMMARY

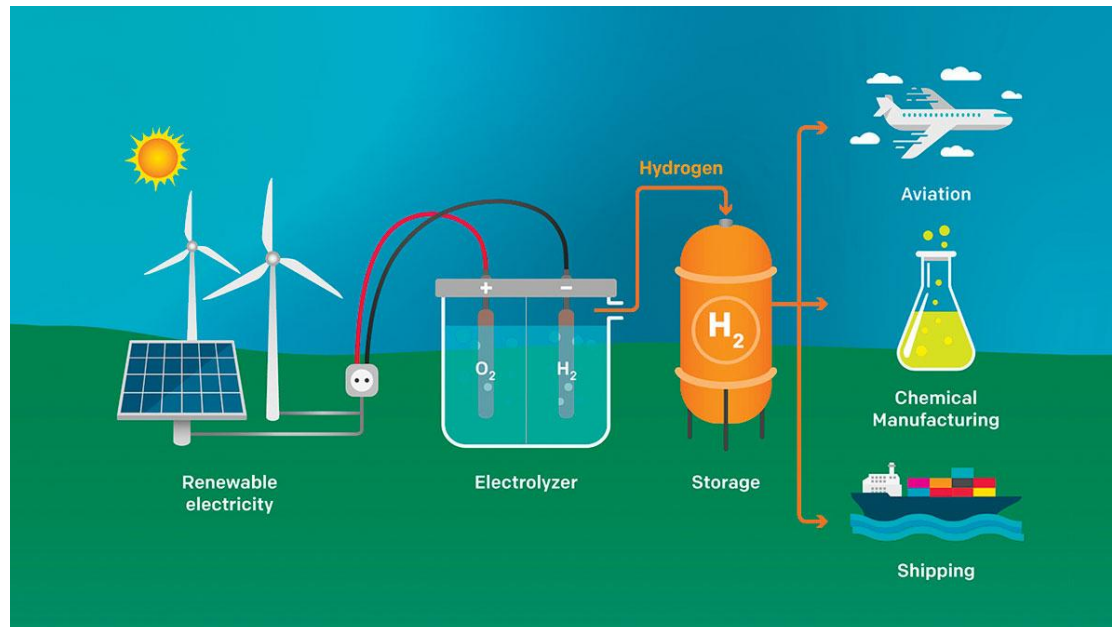
- **Global demand:** Investors are eager for stable oil and LNG sources amid Middle East conflicts, Russia's war in Ukraine, and Venezuela's isolation.
- **Añelo boom:** Fracking transformed a sleepy town into a bustling oil hub.
- **Shale surge:** Vaca Muerta leads a record-breaking rise in shale output and investment.
- **Growing pains:** Roads, housing, and services can't keep up with the pace.
- **Market pivot:** Milei's pro-market reforms and incentives are unlocking fresh capital.
- **Export hurdle:** Oil moves fast; gas face bottlenecks from a lack of LNG infrastructure.
- **Big ambitions:** Vaca Muerta could rival Argentina's farm exports by 2030.
- **Money flowing:** Investment may hit \$23B/year—if logistics can catch up.



HYDROGEN

What is green hydrogen?

- Made by splitting water (electrolysis), using renewable electricity.
→ No GHG emissions



Blue hydrogen:

- Hydrogen from natural gas (reforming)
→ GHG emissions, but captured

Grey hydrogen:

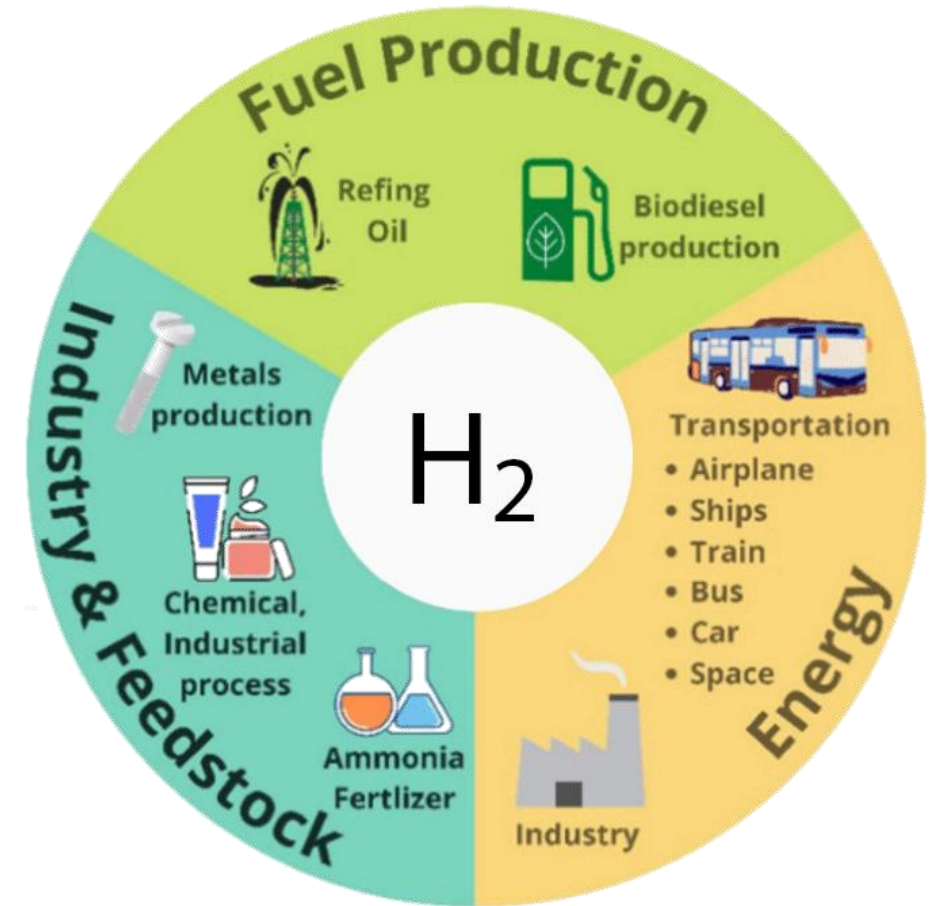
- Hydrogen from natural gas (reforming)
→ GHG emissions, NOT captured

Brown and black hydrogen:

- Hydrogen from brown and black coal
 - Via gasification → emissions

WHY HYDROGEN?

- Zero-emission fuel!
- Stores excess renewable energy: Converts solar/wind surplus into storable, transportable fuel.
- Enables clean industry: Powers green steel, ammonia, and heavy transport where batteries fall short.
- Global export potential: Allows countries with rich renewables (like Chile) to “export sunshine” as fuel.



lenntech.com

CHILE'S HYDROGEN PLAN

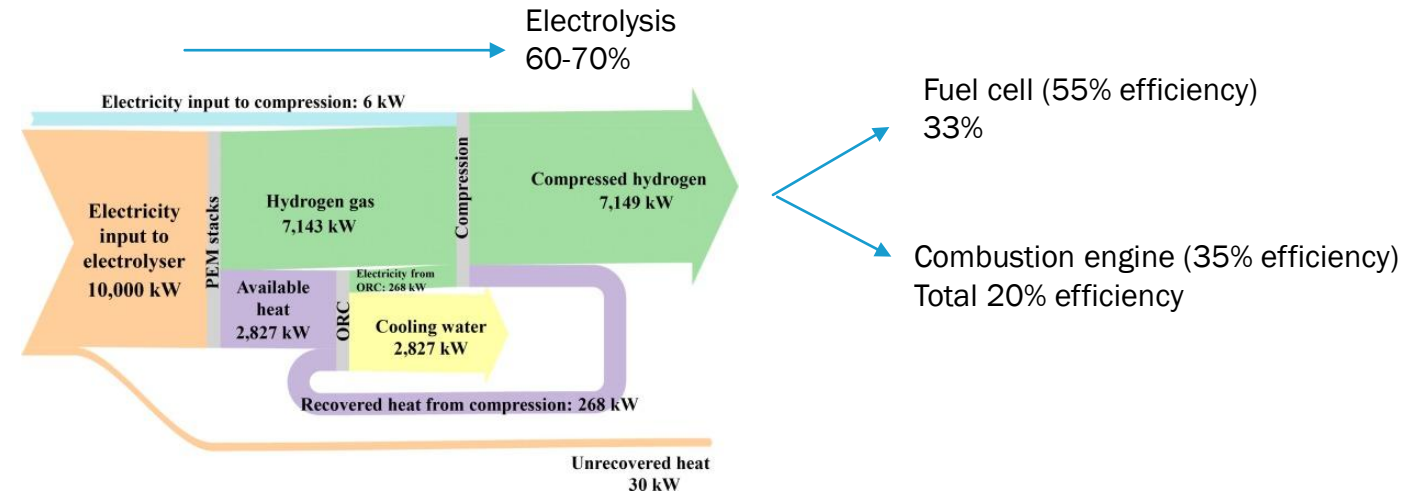
- Ambitious Targets: 25 GW of electrolysis by 2030 and world's cheapest green H₂ (<US\$1.5/kg). <https://gh2.org/countries/chile>
- Export Focus: Aims to ship green ammonia and e-fuels to Europe, Japan, and beyond.
- Major Hurdles: Slow permits, lack of infrastructure, and water scarcity in key regions.
- Losing Ground: Brazil and Australia moving faster with stronger support and investment.



“...posicionarnos como uno de los productores más competitivos a nivel mundial, siendo de los principales exportadores de aquí al 2040.”

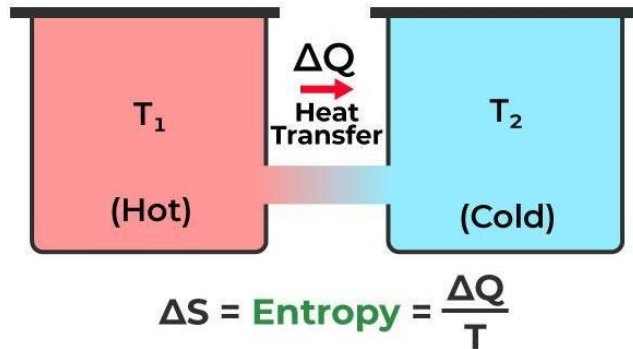
REALITY CHECK

- Hydrogen is not a source!
 - It's an energy carrier that always requires more energy to produce than it delivers.
- Low efficiency – Electrolysis + conversion losses mean only ~20-40% of the original energy is usable.



<https://www.sciencedirect.com/science/article/pii/S0306261923006979>

2nd Law the Thermodynamics



Energy carriers - efficiency

- Direct electricity (battery EV): ~90%
- Gasoline: ~30%
- Green hydrogen (fuel cell): ~30–35%
- Hydrogen combustion: ~20–25%

EROI

- Oil: 10–30
- Natural gas: 10–20
- Solar PV: ~10
- Green hydrogen: ~0.5–0.7

REALITY CHECK

- 2nd Law the Thermodynamics ☒
 - a hydrogen-based energy system cannot sustain itself without ongoing subsidies
- High input, uncertain return – Massive renewable capacity, infrastructure, and subsidies needed for limited energy output.
- Too bold, too fast? – Chile's aggressive targets may outpace what's technically, economically, and socially feasible.



KEY TAKEAWAYS

- **Fossil fuels** still dominate the global energy mix
- **Transition** is real, but slow and uneven
- **Petrochemicals** will remain a key driver of oil demand
- **Markets, policy, and geopolitics** are deeply intertwined

Credits

- Lyn Alden — @LynAldenContact
- Anas Alhajji — @anasalhajji





alan@crowdfield.net
www.crowdfield.net